

Supplementary Appendix

This appendix has been provided by the authors to give readers additional information about their work.

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Supplementary Appendix

Brain health from sleep EEG: A multi-cohort, deep learning biomarker for cognition, disease and mortality

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Table of Contents

Extended main results	3
Methodology.....	6
Data collection – cohort description	6
Sleep feature computation	6
Cognitive test groupings	7
Sleep – Cognition and Sleep – Disease variables associations	7
Deep Learning – Sleep EEG brain health latent space model architecture	8
Deep learning – specific model architectures tested	15
Model evaluation	16
Evaluation metrics for predictive models	16
Model comparisons and additional analyses	17
Sleep – brain health associations	19
Deep learning model architecture and hyperparameter performance comparison	20
Extended model comparison	24
Survival analysis.....	27
Methodology	27
MrOS survival analysis (n = 2,454)	29
FHS survival analysis (n = 744)	29
MGH survival analysis (n = 8,285)	30
BIDMC survival analysis (n = 6,483)	30
Comparison of hazard ratios between brain health score and conventional EEG or cognitive metrics ..	30
Comparison Between Sleep Brain Health Latent Space and EEG-Based Brain Age Index... 32	
Results survival analysis BHS vs. BAI	35
Threshold-based performance analysis	37
Brain health score – age association	43
Sensitivity Analysis of the Brain Health Score Optimization Weight (α).....	45
Sample spectrograms and error analysis	47
Auxiliary tasks results.....	49
Sleep metrics prediction	50
Ablation Without Auxiliary Tasks	74

Extended main results

This section contains figures that directly support the primary findings reported in the main text but are shown here for space and clarity.

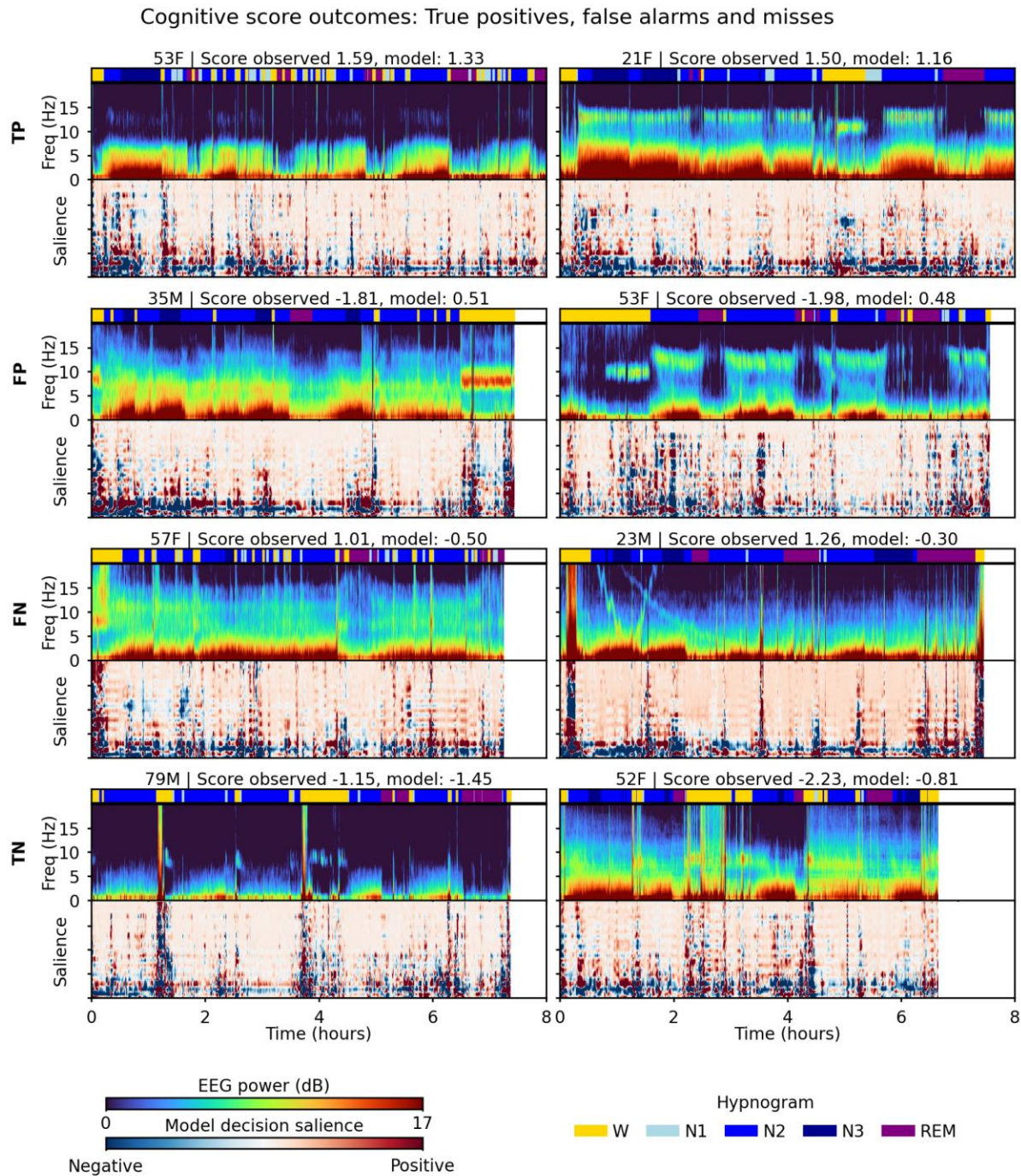


Figure S1: Case-level spectral and saliency patterns underlying cognitive score predictions: Representative examples illustrating the model’s predictive performance for sampled individuals with cognitive scores exceeding one standard deviation from the mean. Each row shows a different outcome category (“true positives (TP),” “false positives (FP),” “false negatives (FN),” and “true negatives (TN)”), with the spectrogram depicting EEG power over time and the bottom panel displaying the Integrated Gradients saliency map (blue-to-red) highlighting features most influential for the model’s predictions. The hypnogram tracks sleep stage transitions (W, N1, N2, N3, and REM). Red salience regions indicate positive contributions toward the predicted cognitive score, whereas blue regions suggest negative contributions. Participants with high observed and predicted cognitive scores typically show clear NREM–REM cycling, strong early-night delta power, and prominent spindle activity, whereas true negatives often exhibit reduced slow-wave and spindle activity and alpha intrusion. False positives and false negatives display less distinct spectral patterns, suggesting more complex individual variability, highlighting potential for further analysis and model refinement.

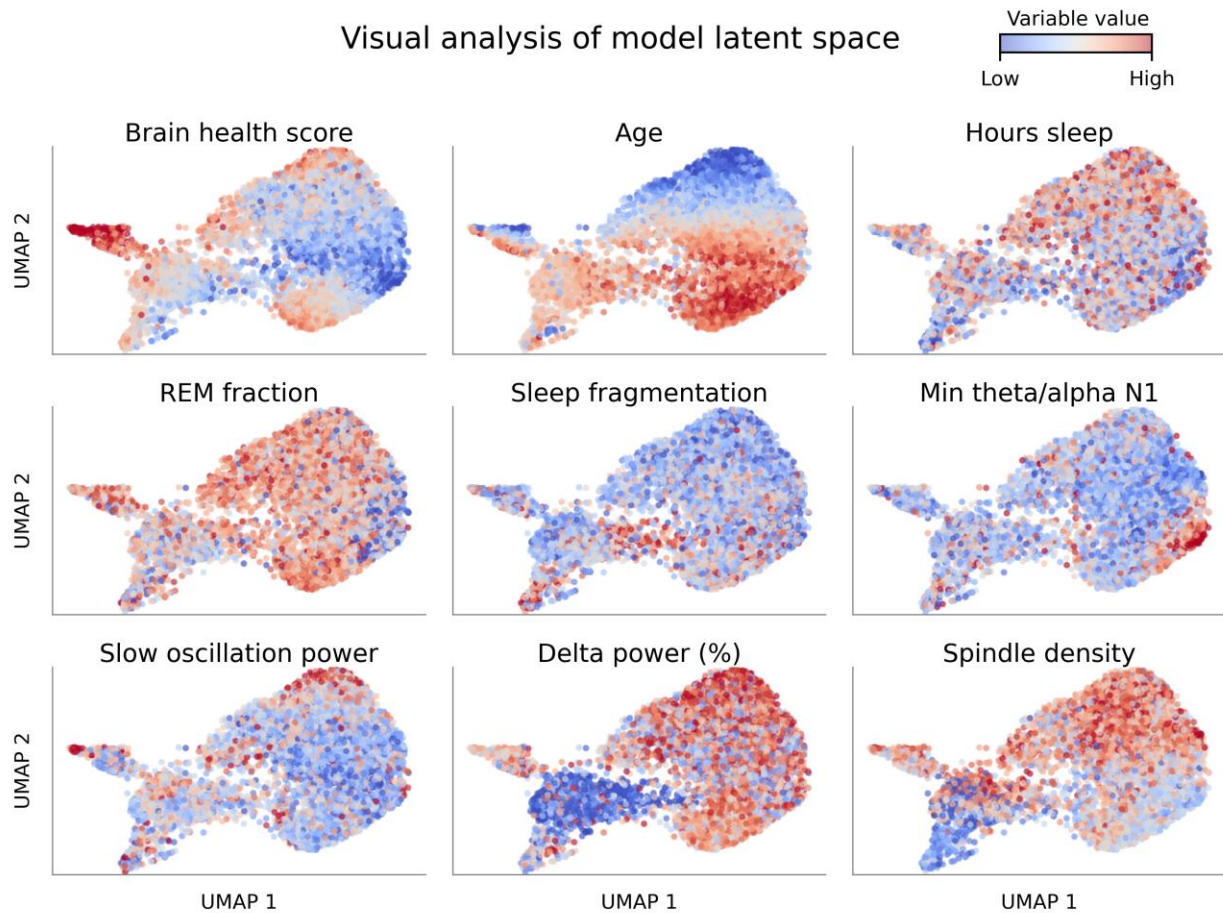


Figure S2. UMAP visualization of the model’s high-dimensional latent space reveals multidimensional sleep EEG patterns. Each panel shows the same 2D UMAP (unsupervised) projection of the model’s 1024-dimensional latent brain health embeddings, with points colored by the indicated variable (blue = low, red = high). The model’s brain health score, age, total hours slept, REM fraction, sleep fragmentation, minimum theta/alpha in N1, slow oscillation power, delta power, and spindle density all exhibit distinct gradients in this shared embedding. The coherent organization across known physiological metrics and newly derived features (brain health score) underscores the network’s capacity to encode rich neurophysiological information. This representation can provide deeper insights into individual brain health profiles, enabling more personalized diagnostics and interventions.

Methodology

Data collection – cohort description

- 1) The Massachusetts General Hospital's Cognitive cohort (MGH-COG): participants were scheduled for clinical PSG consented to complete the NIH Toolbox cognitive battery within 30 days (N=189, mean age 51 ± 17 years, 60% female).
- 2) MGH-MMSE: a retrospective cohort comprising patients who underwent clinical polysomnography and had Mini-Mental State Examination scores available (N=158, mean age 66 ± 11 years, 47% female).
- 3) Framingham Heart Study (FHS): with in-home PSG and cognitive tests conducted between 1995-2003 (N=1319, mean age 60 ± 10 years, 53% female).
- 4) Multi-Ethnic Study of Atherosclerosis (MESA: with in-home PSG and cognitive tests conducted between 2010–2012 (N=1,751, mean age 68 ± 9 years, 54% female).
- 5) MrOS (Osteoporotic Fractures in Men): a large-scale study of older men conducted between 2003–2005 (N=3,770, mean age 77 ± 6 years, 0% female).
- 6) SOF (Study of Osteoporotic Fractures): enrolled elderly women between 2002–2004 (N=426, mean age 83 ± 3 years, 100% female) (Stone et al., 2009).
- 7) KoGES (Korean Genome and Epidemiology Study), involving overnight PSG and comprehensive neuropsychological assessments adapted from the Framingham Heart Study (N=6,224, mean age 62 ± 7 years, 50% female).
- 8) MGH-General cohort: consists of patients who underwent clinical polysomnography at Massachusetts General Hospital with electronic health records but without standardized cognitive testing (N=2,669; mean age 60 ± 13 years, 50% female). This dataset provided disease-status labels (e.g., dementia, cardiovascular diseases, diabetes, psychiatric disorders) extracted from diagnostic codes in electronic health records.

Sleep feature computation

For each stage (N1, N2, N3, REM, W), frequency-specific bandpowers—Slow (0.4–1 Hz), Delta (1–4 Hz), Slow-Delta (0.4–4 Hz), Theta (4–8 Hz), Alpha (8–12 Hz), Sigma (12–16 Hz), Beta (16–30 Hz), Gamma (30–50 Hz), and Fast (8–50 Hz)—were computed in 2-second windows within each 30-second epoch. These microstructural features were described using the minimum, mean, standard deviation, kurtosis and maximum (for 2-second windows within 30-second epochs), then averaged across all sleep epochs for a given sleep stage. Spindle (SP) metrics include density (number of spindles per minute in N2) for distinct spindle types—Slow SP (central frequency ~ 11.5 Hz), Medium SP (~ 13.5 Hz), and Fast SP (~ 15.5 Hz)—along with integrated spindle activity (ISA). Slow oscillations (SO), defined as events with amplitudes exceeding $70 \mu\text{V}$, were identified in N2 and N3. Spindle–SO coupling was quantified based on the co-occurrence of spindles and SOs in those stages. Also computed were total bandpower fractions by stage and overall sleep duration.

Cognitive test groupings

KoGES

Fluid: Digit Span, Logical Memory, Stroop, Trail Making, Verbal Recall

Crystallized: Verbal Fluency

MrOS

Fluid: Recall, Registration, Reverse, Trail Making B (seconds), Digit Vigilance

Crystallized: Naming, Spatial Orientation, Temporal Orientation

SOF

Fluid: Trail Making Test B, MMSE

Crystallized: SMMS

FHS

Fluid: Logical Memory Delayed, Visual Reproduction Delayed, Paired Associates (Immediate & Delayed), Trail Making Test B

Crystallized: Similarities, Boston Naming Test

MESA

Fluid and crystallized cognition were not distinguished in MESA, given the limited specificity of the available tests.

MGH-COG

Fluid: Working Memory, Attention, Executive Function, Processing Speed

Crystallized: Vocabulary, Reading

Sleep – Cognition and Sleep – Disease variables associations

We performed the following analyses: (1) Pearson Correlation: We calculated Pearson correlation coefficients between all sleep and cognitive features to assess linear associations; (2) Partial Pearson Correlation: To account for potential confounding variables, we computed partial Pearson correlations between all sleep and cognitive features, adjusting for age and sex; (3) Maximal Information Coefficient (MIC): Recognizing that relationships between variables may be non-linear, we employed the Maximal Information Coefficient to capture both linear and non-linear associations between all sleep and cognitive features.

To examine relationships between sleep and disease, we conducted analyses comparing each disease category to demographic-matched controls. Specifically, we calculated Cohen's d to quantify effect sizes of differences in sleep variables between groups with and without the disease. Additionally, we employed paired two-sample t-tests to assess statistical significance between disease-positive and control subjects.

Deep Learning – Sleep EEG brain health latent space model architecture

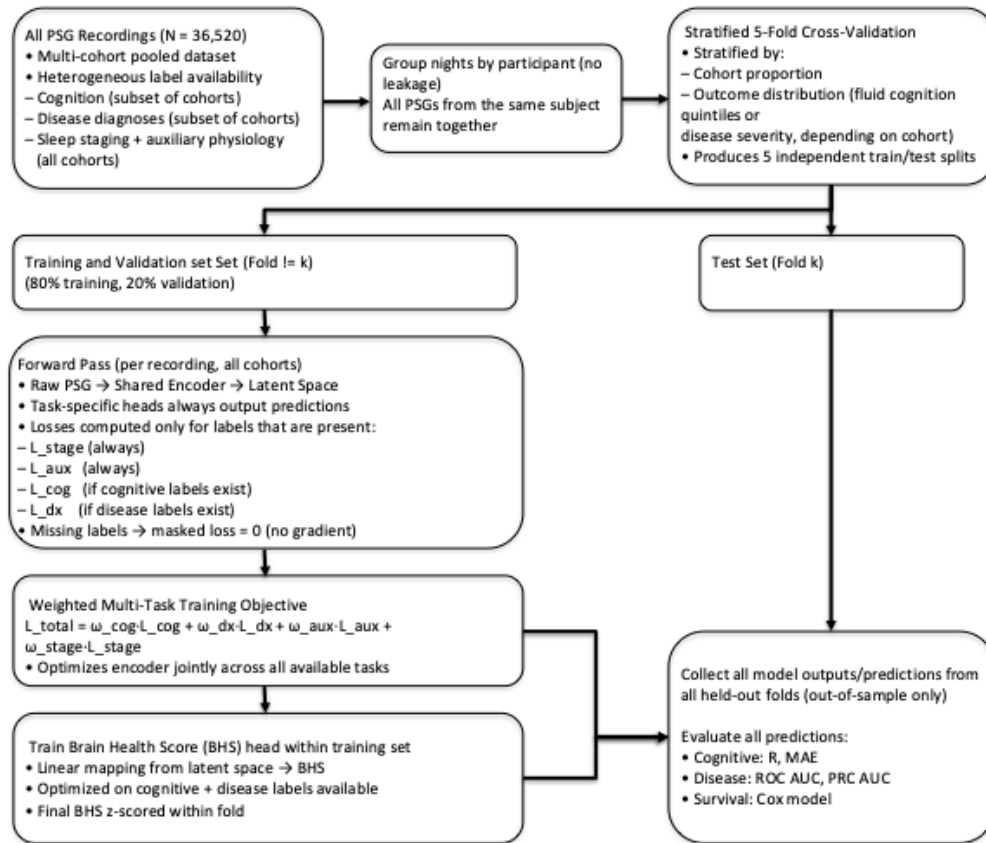


Figure S3. Data splitting, label availability, and multi-task training workflow.

Overview

We developed two main architectures to map overnight EEG data and optional covariates (e.g., demographics) into four prediction targets: (1) cognition regression, (2) disease classification, (3) sleep-stage segmentation, and (4) self-supervised-like sleep feature regression. Both architectures share a high-level design: an encoder (plus a bottleneck) produces a compact latent “brain-health” representation, which then branches into two decoding strands:

1. Transformer-based head for cognition, disease, and self-supervised tasks
2. U-Net-like upsampler with skip connections for sleep-staging outputs

However, the specifics of data ingestion, transformation, and downsampling differ for:

(i) Spectrogram inputs (2D): The model receives wavelet or multitaper time × frequency spectrograms. We experimented with different spectrogram time resolutions (step sizes 1, 2, 4, 8 seconds, with 1 and 2 seconds showing best performance, we selected stepsize 1 second); sleep EEG was zero-padded to 11 hours, the final spectrogram size was (39,600 × 100), with 39,600

one-second timesteps for 11 hours, and 100 frequency bins. We employ MaxVit blocks—an efficient Transformer design that factorizes attention along spatial (frequency/time) dimensions to handle large images without quadratic memory overhead. A series of MaxVit stages progressively downsample the spectrogram, capturing long-range dependencies. The final encoder output is distilled into a 1024-dimensional latent bottleneck. A Transformer head processes the bottleneck to predict cognition/disease/self-supervised outputs, while a U-Net–style decoder—using skip connections from each MaxVit stage—upsamples the representation to produce multi-class sleep-stage predictions.

(ii) Time-series inputs (1D): This model takes 1D EEG signals at 200 Hz (792,000 points per 11-hour night). A CNN stem with multiple 1D convolutional blocks efficiently reduces the time points to a smaller feature map, ultimately producing a manageable number of tokens (387 for the final model) for the Transformer layers. Each attention block includes depthwise-separable convolutions to further reduce temporal resolution, and a positional encoding preserves ordering. Similar to the spectrogram variant, the encoder culminates in a 1024-dimensional latent representation. From here, a Transformer head outputs cognition/disease/self-supervised predictions, and a U-Net–style decoder reconstructs 1D feature maps for sleep-stage classification, leveraging skip connections from both the CNN stem and the Transformer layers.

We tested model configurations ranging from 20 million to 250 million parameters, varying by the number of Transformer stages, convolutional layers and strides, patch sizes, and attention parameters.

Loss function: In the following equations, the superscript (i) denotes an input sample, i.e. a full sleep EEG recording. Each loss component is calculated per sleep recording before being aggregated during training.

Regression tasks (cognition, self-supervised sleep features): $\mathcal{L}_{reg}^{(i)} = \mathcal{L}_{Huber}(\hat{y}^{(i)}, y^{(i)})$

Classification tasks (diseases): $\mathcal{L}_{cls}^{(i)} = [y^{(i)} \log(\hat{p}^{(i)}) + (1 - y^{(i)}) \log(1 - \hat{p}^{(i)})]$

Sleep staging: $\mathcal{L}_{stg}^{(i)} = \frac{1}{2} \mathcal{L}_{CE}(\hat{s}^{(i)}, s^{(i)}) + \frac{1}{2} \mathcal{L}_{dice}(\hat{s}^{(i)}, s^{(i)})$

Total weighted loss: $Loss_{total}^{(i)} = [1 - \alpha_1 - \alpha_2] \mathcal{L}_{reg}^{(i)} + \alpha_1 \mathcal{L}_{cls}^{(i)} + \alpha_2 \mathcal{L}_{stg}^{(i)}$

In addition to training separate heads for each cognition and disease task, we introduce a single linear aggregator of the 1024 latent space that distills these task outputs into a single continuous metric we term the ‘brain health score.’ This aggregator is trained with a joint objective that combines Huber regression loss for cognitive tasks and inverted-logit classification loss for disease tasks, thereby aligning higher brain health score values with better cognitive performance and lower disease probability. In effect, the network learns a unified latent index of overall ‘brain health’ that balances multiple task-specific objectives simultaneously.

Training and Implementation: Both models were implemented using PyTorch Lightning (v2.0.1) and optimized with AdamW (learning rate: 1e-4, weight decay: 2e-4). Training was conducted with PyTorch (v2.0.0) and CUDA Toolkit (v11.6.0).

Detailed description

Additional, more specific information to the models used (specifically v2_3 which performed best in the spectrogram input case, and 3_6LA1 which performed best in the timeseries input case) are presented below.

Brain health latent space model with spectrogram inputs (model v2_3):

1. CNN Stem:

- a. Design: The model begins with a convolutional “stem” that is sensitive to the sampling frequency (fs_time). Depending on whether fs_time is 1, 2, 4, or 8, the stem conditionally adds extra pre-convolution layers to adjust the input’s temporal resolution.
- b. Operations: A series of 2D convolutional layers (with chosen kernel sizes, strides, and paddings) extract local spatial and temporal features from the input. Batch normalization (or alternative normalization layers) and non-linear activations (e.g., GELU) follow each convolution.
- c. Intuition: This part acts as a feature extractor, “pre-processing” the raw sleep EEG signal into a representation that can be meaningfully partitioned into tokens for subsequent transformer operations.

2. Transformer-Based Stages (MaxxVit Stages):

- a. Design: After the CNN stem, the model passes the intermediate features through several MaxxVit stages. These stages integrate both convolutional and transformer blocks.
- b. Operations: Each stage employs multi-head self-attention (with relative positional encoding via MLPs) alongside convolutional sub-blocks. This hybrid approach is intended to capture both local (convolutional) and global (attention-based) dependencies.
- c. Hypothesis: The belief here is that the self-attention mechanism will help the model understand long-range interactions in the EEG data.

3. Multi-Task Branching:

- a. Sleep Stages Decoder: A dedicated decoder network (built primarily with ConvTranspose2d layers) upsamples the features to a resolution that matches the temporal granularity of 30-second sleep stages. Skip connections from earlier layers in the CNN stem (and possibly from intermediate transformer stages) are concatenated during upsampling. This U-Net-like strategy helps preserve spatial details that might otherwise be lost in deeper layers.
- b. Final Latent Space & Heads: Parallel to the sleep staging branch, the model further processes the high-level features through additional transformer blocks.
- c. Output: From this final latent representation, task-specific linear heads predict the regression outputs (e.g., cognitive scores) and classification labels (e.g., disease states), as well as self-regularized tasks (regression tasks, sleep metrics)

4. Framework choice: The model adapts core model building blocks and operations from the pytorch timm library.

5. Pseudo-code:

```

Function UnifiedBrainHealthModel(input, covariates, fs_time=1):

    # === CNN Stem ===
    If fs_time == 1:
        conv1 = Conv2d(1, 16, kernel_size=(3,2), stride=(2,1))

    Else if fs_time == 2:
        conv_pre1 = Conv2d(1, 16, kernel_size=(3,2), stride=(2,1), padding=(1,0))
        norm_pre1 = NormActLayer(16, eps=1e-5)
        padding_pre1 = ConstantPad2d((0,1,0,0))
        conv1 = Conv2d(16, 16, kernel_size=(3,2), stride=(2,1))

    Else if fs_time in [4, 8]:
        conv_pre1 = ...
        norm_pre1 = ...
        padding_pre1 = ...
        If fs_time >= 4:
            conv_pre2 = Conv2d(16, 16, kernel_size=(3,2), stride=(2,1), padding=(1,0))
            norm_pre2 = NormActLayer(16, eps=1e-5)
            padding_pre2 = ConstantPad2d((0,1,0,0))
        If fs_time == 8:
            conv_pre3 = Conv2d(16, 16, kernel_size=(3,2), stride=(2,1), padding=(1,0))
            norm_pre3 = NormActLayer(16, eps=1e-5)
            padding_pre3 = ConstantPad2d((0,1,0,0))
        conv1 = Conv2d(16, 16, kernel_size=(3,2), stride=(2,1))

    norm1 = NormActLayer(16, eps=1e-5)
    conv2 = Conv2d(16, 32, kernel_size=(3,2), stride=(2,1))
    norm2 = NormActLayer(32, eps=1e-5)
    conv3 = Conv2d(32, 64, kernel_size=(3,2), stride=(2,1))
    norm3 = NormActLayer(64, eps=1e-5)
    conv4 = Conv2d(64, embed_dim, kernel_size=(3,2), stride=(2,2))
    pad_end = ConstantPad2d((0,0,0,1))

    # Forward pass through stem
    x1 = ProcessThroughPreLayers(input)
    x1 = norm1(conv1(x1))
    x2 = norm2(conv2(x1))
    x3 = norm3(conv3(x2))
    x4 = pad_end(conv4(x3))
    stem_outputs = [x1, x2, x3, x4]

    # === MaxxVit Feature Extraction ===
    stages = []
    For i in range(num_stages):
        stage_i = MaxxVitStage(
            in_channels = previous_channels,
            out_channels = cfg.embed_dim[i],
            depth = cfg.depths[i],
            block_type = cfg.block_type[i],
            conv_cfg = cfg.conv_cfg,
            transformer_cfg = AdjustWindowAndGridSize(i),
            feat_size = computed_feature_size,
            drop_path = dpr[i]
        )
        stages.append(stage_i)
        previous_channels = cfg.embed_dim[i]

    x = Sequential(stages)(x4)
    x = NormLayer()(x)

```

```

# === Sleep Stage Decoder (U-Net Style) ===
x_decoder = MaxxVitStage(
    in_channels=n_chs, depth=1, block_types='M',
    conv_cfg=cfg.conv_cfg, transformer_cfg, feat_size, stride=1, drop_path=[0]
)

layer1 = Sequential([
    ConvTranspose2d(n_chs, 512, (3,2), (2,2)), ReLU(), ConstantPad2d((0,0,0,0)),
    ConvTranspose2d(512, 256, (3,2), (2,2)), ReLU(), ConstantPad2d((0,0,0,1)),
    ConvTranspose2d(256, 128, (3,2), (2,2)), ReLU(), ConstantPad2d((0,0,0,1)),
    ConvTranspose2d(128, 64, (3,2), (2,2)), ReLU(), ConstantPad2d((0,1,0,0)),
])

layer2 = ConvTranspose2d(64 * 2, 32, (3,2), (2,1))
layer3 = ConvTranspose2d(32 * 2, 16, (3,2), (2,1))
layer4 = Sequential([
    ConvTranspose2d(16 * 2, 1, (3,2), (2,1)),
    ReLU(), ConstantPad2d((0,0,0,1)),
])

If 30 % fs_time == 0:
    layer_final = Conv2d(1, 6, (30 // fs_time, 100), stride=(30 // fs_time))
Else:
    layer_final = Sequential([
        Conv2d(1, 6, (30 // fs_time, 100), stride=(30 // fs_time)),
        SliceModule(target_length=1320 * fs_time)
    ])

yp_stages = layer_final(layer4(layer3(layer2(layer1(x_decoder + skip_connections)))))

# === Regression & Classification ===
features_lhl = final_latent_space_mlp(regression_vit(x), covariates)

For each head in heads:
    output[head] = Linear(features_lhl)

yp_regression = Concat(outputs for regression targets)
yp_classification = Concat(outputs for classification targets)

Return yp_regression, yp_classification, yp_stages

```

Brain health latent space model with timeseries inputs (model v3_6L):

1. 1D CNN Stem (Stem1D):

- a. **Design:** Instead of processing 2D “images” (like in v2_3), this model employs a 1D convolutional stem that directly works on the time-series data.
- b. **Operations:** Three sequential Conv1d layers (each followed by a normalization and a residual connection) are used to extract features at multiple resolutions. The design is reminiscent of classic deep CNNs, but with residual shortcuts that ease the training of deep networks.

- c. **Role:** This stem reduces the temporal dimensionality and produces a set of feature maps that capture localized patterns in the EEG signals.
2. **Transformer Block with Convolutional Augmentation:**
 - a. **Design:** After the stem, the model transforms the feature maps (by permuting the tensor dimensions) into a sequence of tokens suitable for transformer processing.
 - b. **Positional Encoding:** A standard positional encoding module is applied to inject order information.
 - c. **Custom Attention-Conv Block:** This block comprises multiple layers where each layer starts with multi-head self-attention (to capture global dependencies) followed by a feedforward network. Between layers, depthwise separable convolutions are used, which introduces local inductive biases back into the sequence model. Skip connections from each layer are stored and later used in the decoder, ensuring that information from different depths is retained.
3. **Dimensionality Reduction & Latent Representation:** A specialized module applies a residual block combined with attention pooling to collapse the high-dimensional sequence into a compact last hidden layer latent representation.
4. **Sleep Staging Decoder (U-Net-Style):** A decoder inspired by U-Net architectures is used to reconstruct the sleep stage predictions from the latent features.
 - a. **Upsampling:** The decoder uses a series of ConvTranspose1d layers to progressively upsample the feature maps.
 - b. **Skip Connections:** It concatenates skip connections (coming from both the CNN stem and the transformer block) to preserve high-resolution details.
 - c. **Final Convolution:** A concluding Conv1d layer maps the upsampled features to the desired number of sleep stage classes.
 - d. **Output Resolution:** The final output aligns with the temporal resolution required for sleep staging (typically one prediction per 30-second epoch).
5. **Task-Specific Heads:** In parallel with the sleep staging decoder, the model also includes a set of linear heads for regression and classification. These heads operate on the final latent representation produced after the dimensionality reduction stage.
6. **Software & Framework Choices:** Like the v2_3 model, v3_6L is built on top of a shared base class (BaseOracleModule), ensuring consistency in training and evaluation routines.
7. **Framework choice:** The implementation makes use of PyTorch modules and custom layers.
8. **Pseudocode:**

```
UnifiedBrainHealthModel_v3_6L(input, covariates, fs_time=200, seq_length=792000):
    # === Input Normalization ===
    x = BatchNorm1d(input) // Normalizes raw 1D EEG signal

    # === 1D CNN Stem (Stem1D) ===
    # Three sequential Conv1d layers with residual connections:
    # Parameters for Stem1D:
    # Kernel sizes: [100, 25, 16]
    # Strides: [75, 21, 13]
    # Paddings: [49, 13, 7]
    conv1 = Conv1d(in_channels=1, out_channels=128, kernel_size=100, stride=75, padding=49)
    norm1 = NormActLayer(128)
```

```

residual1 = (If input channels != 128 then Conv1d(1,128, kernel_size=1, stride=75))
x1 = norm1(conv1(x) + residual1(x))
Store x1 in skip_connections_stem

conv2 = Conv1d(128, 512, kernel_size=25, stride=21, padding=13)
norm2 = NormActLayer(512)
residual2 = (If 128 != 512 then Conv1d(128,512, kernel_size=1, stride=21))
x2 = norm2(conv2(x1) + residual2(x1))
Store x2 in skip_connections_stem

conv3 = Conv1d(512, 1024, kernel_size=16, stride=13, padding=7)
norm3 = NormActLayer(1024)
residual3 = (If 512 != 1024 then Conv1d(512,1024, kernel_size=1, stride=13))
x3 = norm3(conv3(x2) + residual3(x2))
Store x3 in skip_connections_stem
# Stem1D outputs: x = x3, and skip_connections_stem = [x1, x2, x3]

# === Transformer Block with Convolutional Augmentation ===
# Permute x to (batch, sequence_length, embed_dim)
x = Permute(x3, to_shape=(batch, L, 1024))
x = PositionalEncoding(embed_dim=1024, max_len=cnn_out_length)(x)
# Custom Attention-Conv Block with N layers (e.g., N=5)
For layer_idx in 1 to N:
    If pre-layer normalization enabled:
        x = LayerNorm(x)
    # Multi-Head Self-Attention:
    attn_output = MultiHeadAttention(embed_dim=1024, num_heads=(1024/dim_attention_head))(x, x, x)
    x = x + attn_output // Residual connection
    # Feedforward Network:
    ffn_output = FFN(x) // e.g., Linear(1024,4096) -> GELU -> Linear(4096,1024)
    x = x + ffn_output
    # Depthwise Separable Conv1d (applied on transposed tensor):
    x_conv = Transpose(x, to_shape=(batch, 1024, L))
    x_conv = DepthwiseSeparableConv1d(1024, 1024, kernel_size=3, stride=2, padding=1)(x_conv)
    x = Transpose(x_conv, to_shape=(batch, L_new, 1024))
    Store skip connection from current layer (after conv) in skip_connections_attention_conv

# === Combine Skip Connections ===
skip_connections = skip_connections_stem + skip_connections_attention_conv

# === Sleep Staging Decoder (U-Net Style for 1D Data) ===
# Permute x to (batch, channels, L)
x = Permute(x, to_shape=(batch, 1024, L_new))
# Decoder consists of a series of ConvTranspose1d layers.
For each skip connection in reverse order:
    x = ConvTranspose1d(x, out_channels=target, kernel_size=decoder_kernel, stride=decoder_stride,
padding=decoder_padding)
    x = ReLU(x)
    # Adjust dimensions if necessary to match the skip connection size:
    x = x[:, :, :size_of(skip_connection)]
    x = Concat(x, skip_connection) along channel dimension
# After processing all skip connections:
x = one_second_output_block(x)
// e.g., Conv1d(kernel_size=5, stride=5) -> ReLU -> Conv1d(kernel_size=16, stride=16) -> ReLU
yp_stages = final_conv(x) // Final Conv1d: maps channels to 6 sleep stage classes

# === Latent Feature Extraction & Task-Specific Heads ===
# Enhanced Dimensionality Reduction:
features = EnhancedDimReduction(
    ResidualBlock1d(1024, dim_final_latent_space),

```



```

        AttentionPooling1d(dim_final_latent_space, num_heads=8)
    )(x)

```

```

For each target in regression/classification tasks:
    output[target] = Linear(features, output_dim_for_target)
yp_regression = Concat(regression outputs)
yp_classification = Concat(classification outputs)

```

```

Return yp_regression, yp_classification, yp_stages

```

Both models were designed to adapt recent methodologies in deep learning by blending convolutional layers (for local feature extraction) with transformer-based attention mechanisms (for capturing long-range dependencies). The v2_3 model is more oriented toward 2D representations of the input, treating the sleep EEG data as high-resolution images (or spectrogram-like representations), whereas the v3_6L model takes a more native 1D approach to time-series data.

Deep learning – specific model architectures tested

Models using spectrogram as input (v2 models):

Model	Params (M)	Δ vs v2_3	Key architectural difference (vs v2_3)
v2_3 (ref)	199.3	0%	12 transformer blocks; 384-dim embedding; 8 attention heads; standard MLP (1536 hidden)
v2_1	200.2	+0.5%	Same depth; embed $\uparrow$ to 400; same heads; identical block structure
v2_2	118.2	-40.7%	Reduced to 8 transformer blocks; embed $\downarrow$ to 256; 6 heads; MLP hidden dim $\downarrow$ to 1024
v2_3D2	200.1	+0.4%	Same as v2_3 but adds dropout ($p=0.1$) after each MLP
v2_3D3	200.1	+0.4%	Same depth & width; MLP hidden dim $\uparrow$ to 2048
v2_4	247.7	+24.3%	Depth $\uparrow$ to 16 blocks; embed $\uparrow$ to 512; 12 heads; MLP hidden dim $\uparrow$ to 2048
v2_5	164.4	-17.5%	Same 12 blocks; embed $\downarrow$ to 320; 6 heads; MLP hidden dim $\downarrow$ to 1280
v2_6	179.3	-10.0%	Depth $\downarrow$ to 10 blocks; embed =384; 8 heads
v2_7	161.6	-18.9%	Same 10 blocks; embed $\downarrow$ to 300; 6 heads
v2_8	247.6	+24.2%	Identical to v2_4 (16 blocks, 512-dim), but MLP hidden dim $\uparrow$ to 3072
v2_9	200.2	+0.5%	Duplicate of v2_1

Models using timeseries as input (v3 models):

Famil y	Param s (M) range	Δ vs ref (%)	Conv-ste m width (channels)	LA block s	Embed dim (A variant s)	Embed dim (LA variants)	Key architectural difference (vs v3_6LA1)
v3_0	17.4 → 158.5	−74.2 → +135.7	64 (6 conv layers)	8	128→ 256→ 320	384→5 12→76 8	Smallest conv-stem; LA adds 8 blocks; embed dims scale from 128→768
v3_1	22.2 → 162.7	−67.1 → +141.8	96	8	192→ 320→ 448	384→7 68→10 24	Moderate conv width; LA introduces full attention stack
v3_2	16.5 → 140.1	−75.4 → +108.2	80	8	128→ 256→ 384	384→7 68→10 24	Narrow conv stem; LA adds same 8 blocks
v3_3	21.5 → 160.0	−68.1 → +137.9	96	8	192→ 320→ 448	384→7 68→10 24	Similar to v3_1 but larger param footprint
v3_4	34.2 → 160.6	−49.2 → +138.6	128	8	256→ 384→ 512	384→7 68→10 24	Wider conv stem; LA dims match reference
v3_5	41.5 → 189.9 7	−38.3 → +182.3	160	8	320→ 448→ 576	512→7 68→10 24	High-capacity stem; largest LA variant is +182% params
v3_6	33.3 → 159.7	−50.6 → +137.4	192	8	384→ 512→ 640→ 768	384→7 68→10 24	

Model evaluation

For classic ML experiments, we use 10-fold nested cross-validation (CV), while for deep learning (DL), we apply 5-fold CV. To prevent information leakage, test folds remain strictly independent, ensuring that no data from them influences model training. Final model outputs are derived exclusively from these held-out test folds. Additionally, for subjects with multiple PSG recordings, all recordings are kept within the same fold to maintain consistency.

Evaluation metrics for predictive models

To assess cognitive task performance, we utilize linear regression-based models, including final layers in deep neural networks, to capture the continuous nature of cognitive test scores. Model predictions, obtained from test folds (see below), are evaluated using two key metrics: (1) Pearson correlation, which measures the linear relationship between predicted and observed cognitive scores, and (2) Median absolute error (MAE), which quantifies the typical prediction error while being robust to outliers.

To assess the performance of disease classification models, we employ logistic regression-based models, including final layers in deep neural networks, to distinguish between “disease” and matched “control” groups. Predictive performance is evaluated using two key metrics: (1) Area Under the Receiver Operating Characteristic Curve (AUC-ROC): Evaluates model performance by trading off sensitivity and specificity across classification thresholds; (2) Area Under the Precision-Recall Curve (AUC-PRC): Assesses predictive performance by balancing precision and sensitivity (recall).

For all metrics, we compute 95% confidence intervals by bootstrapping the model outputs 10,000 times, resampling with replacement.

Model comparisons and additional analyses

To compare models, we evaluated differences in R, MAE, ROC AUC, and PRC AUC using 95% confidence intervals.

We further stratified participants by age and examined the discriminative validity of the brain health score by comparing: (1) subjects in the lowest versus highest quartiles of observed cognitive scores; (2) disease-free controls versus those with diagnoses of mild cognitive impairment (MCI), dementia, depression, or bipolar disorder. Group differences were tested using two-sample t-tests. Additionally, we quantified the proportion of variance in brain health scores explained by age across the entire cohort.

For case-level examination and error analysis, we selected representative participants whose observed cognitive scores exceeded one standard deviation from the mean. Within this subgroup, we identified subjects corresponding to extremes of model predictions, facilitating identification of true positives, false positives, false negatives, and true negatives.

To interpret neural network decision-making, we applied Integrated Gradients, a saliency mapping technique that quantifies the contribution of parts of the original sleep data input to model predictions, visualized via a standardized blue-to-red gradient representing negative-to-positive activation values.

Lastly, to visualize the high-dimensional latent representation, we applied unsupervised Uniform Manifold Approximation and Projection (UMAP) to reduce the 1024-dimensional embeddings to two dimensions. UMAP embeddings were visualized in relation to the derived brain health score and several key sleep variables, including age, total sleep duration, REM fraction, sleep fragmentation, minimum theta/alpha power fraction, slow oscillation power (<1 Hz), delta power (1–4 Hz), and sleep spindle density.

Sleep – brain health associations

Associations sleep - cognitive health (unadjusted)

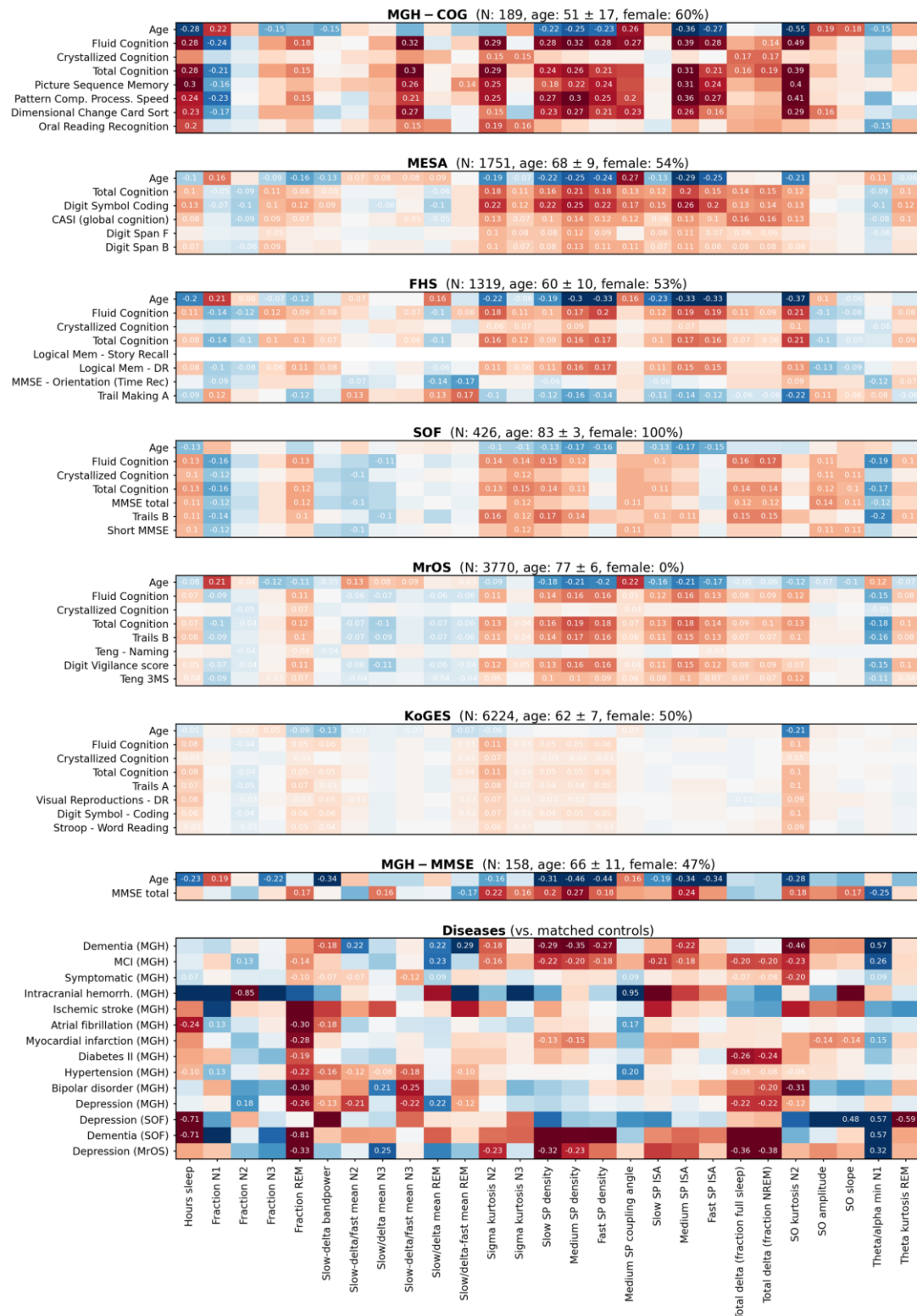


Figure S4 Unadjusted associations sleep – cognition and sleep – diseases

Deep learning model architecture and hyperparameter performance comparison

Table S1 a) Models using spectrograms as input

Model name	Model Type	Input Fs	Model parameters (in M)	Summary performance across tasks (validation)	Summary performance (training)	Learning rate	Validation Cohen's Kappa Sleep staging	Average correlation performance cognitive tests	Mean ROC across diseases	Mean correlation performance or sleep metrics	Performance fluid cognition in MGH	ROC dementia in MGH
v2_3	wavelet	1	199	0.49	0.619	0.0001	0.64	0.276	0.703	0.57	0.61	0.729
v2_1	wavelet	1	200	0.484	0.59	0.0001	0.63	0.268	0.7	0.558	0.583	0.727
v2_1	wavelet	1	200	0.474	0.607	0.00015	0.63	0.269	0.679	0.557	0.619	0.691
v2_3	multitaper	1	199	0.473	0.656	0.0001	0.75	0.269	0.677	0.613	0.574	0.657
v2_1	wavelet	2	200	0.473	0.648	0.0002	0.64	0.263	0.682	0.583	0.615	0.666
v2_1	wavelet	1	200	0.469	0.689	0.0001	0.63	0.261	0.677	0.568	0.62	0.719
v2_3 D3	wavelet	1	200	0.468	0.69	0.0001	0.65	0.261	0.674	0.599	0.605	0.716
v2_3 D2	wavelet	1	200	0.466	0.66	0.0001	0.65	0.265	0.668	0.599	0.67	0.634
v2_4	multitaper	1	248	0.465	0.582	0.0001	0.74	0.254	0.676	0.444	0.596	0.673
v2_1	wavelet	2	200	0.464	0.68	0.0001	0.63	0.252	0.675	0.597	0.573	0.664
v2_1	wavelet	2	200	0.463	0.634	0.0001	0.63	0.259	0.667	0.578	0.545	0.666
v2_1	wavelet	2	200	0.461	0.698	0.0002	0.65	0.242	0.68	0.413	0.626	0.678
v2_1	wavelet	1	200	0.453	0.516	0.0002	0.63	0.262	0.644	0.493	0.549	0.63
v2_1	wavelet	1	200	0.45	0.565	0.0001	0.62	0.245	0.655	0.493	0.606	0.615
v2_1	wavelet	2	200	0.448	0.843	5E-05	0.64	0.243	0.654	0.62	0.611	0.696
v2_4	wavelet	2	248	0.446	0.519	0.0001	0.63	0.236	0.656	0.48	0.526	0.652
v2_4	wavelet	4	248	0.445	0.522	0.0001	0.58	0.24	0.65	0.365	0.516	0.637
v2_3	multitaper	1	199	0.444	0.544	0.0001	0.74	0.259	0.628	0.528	0.589	0.644
v2_1	wavelet	4	200	0.444	0.702	0.0001	0.58	0.263	0.625	0.586	0.553	0.628
v2_4	wavelet	4	248	0.442	0.53	0.0001	0.59	0.246	0.637	0.384	0.457	0.644
v2_1	wavelet	2	200	0.438	0.622	0.0002	0.64	0.25	0.626	0.533	0.542	0.665
v2_3	wavelet	1	199	0.438	0.826	0.0001	0.64	0.23	0.647	0.6	0.589	0.67

v2_4	wavelet	2	248	0.436	0.508	0.0001	0.63	0.235	0.637	0.453	0.548	0.65
v2_1	multiscale	1	200	0.434	0.778	0.0001	0.76	0.245	0.623	0.628	0.495	0.654
v2_1	wavelet	2	200	0.434	0.556	0.0001	0.62	0.246	0.622	0.486	0.576	0.656
v2_1	wavelet	2	200	0.433	0.511	0.0001	0.62	0.233	0.633	0.462	0.512	0.644
v2_1	wavelet	4	200	0.432	0.534	0.0001	0.58	0.262	0.603	0.478	0.557	0.586
v2_1	wavelet	2	200	0.431	0.562	5E-05	0.62	0.246	0.616	0.5	0.576	0.609
v2_3	wavelet	1	199	0.428	0.459	0.0001	0.61	0.232	0.623	0.461	0.518	0.634
v2_3	wavelet	1	199	0.427	0.49	0.0001	0.61	0.22	0.634	0.481	0.555	0.638
v2_1	wavelet	4	200	0.427	0.676	0.0001	0.58	0.218	0.636	0.473	0.585	0.572
v2_3	multiscale	1	199	0.425	0.488	0.0001	0.73	0.237	0.613	0.448	0.538	0.583
v2_4	wavelet	2	248	0.424	0.512	0.0001	0.63	0.227	0.622	0.438	0.529	0.628
v2_1	wavelet	2	200	0.424	0.792	0.0001	0.63	0.237	0.612	0.576	0.566	0.59
v2_1	multiscale	1	200	0.423	0.486	0.0001	0.74	0.247	0.599	0.443	0.549	0.583
v2_1	wavelet	2	200	0.422	0.576	2E-05	0.63	0.235	0.608	0.486	0.576	0.573
v2_7	multiscale	1	162	0.422	0.481	0.0001	0.71	0.234	0.611	0.343	0.561	0.617
v2_1	wavelet	1	200	0.422	0.536	0.0001	0.62	0.248	0.597	0.465	0.635	0.567
v2_1	wavelet	4	200	0.422	0.872	5E-05	0.58	0.201	0.644	0.591	0.585	0.651
v2_8	wavelet	1	248	0.422	0.862	0.0001	0.65	0.21	0.634	0.603	0.623	0.613
v2_4	wavelet	1	248	0.422	0.462	0.0001	0.63	0.22	0.624	0.428	0.322	0.621
v2_6	multiscale	1	179	0.419	0.476	0.0001	0.72	0.185	0.653	0.478	0.533	0.643
v2_4	wavelet	1	248	0.418	0.483	0.0001	0.63	0.235	0.601	0.449	0.487	0.576
v2_1	wavelet	2	200	0.416	0.513	0.0001	0.63	0.23	0.603	0.41	0.527	0.596
v2_8	wavelet	1	248	0.413	0.431	0.0001	0.62	0.234	0.592	0.322	0.521	0.641
v2_3	multiscale	1	199	0.413	0.522	0.0001	0.73	0.176	0.65	0.523	0.424	0.667
v2_3	wavelet	2	199	0.409	0.456	0.0001	0.61	0.251	0.567	0.421	0.596	0.587
v2_1	wavelet	2	200	0.406	0.585	2E-05	0.61	0.228	0.583	0.491	0.473	0.602
v2_1	multiscale	1	200	0.406	0.421	0.0001	0.74	0.196	0.615	0.267	0.578	0.63
v2_4	wavelet	1	248	0.406	0.884	0.0001	0.64	0.176	0.636	0.626	0.568	0.653
v2_5	multiscale	1	164	0.406	0.414	0.0001	0.7	0.203	0.608	0.246	0.531	0.619
v2_3	multiscale	1	199	0.402	0.495	0.0001	0.74	0.162	0.641	0.511	0.315	0.651
v2_1	wavelet	2	200	0.395	0.944	2E-05	0.63	0.172	0.618	0.63	0.63	0.649

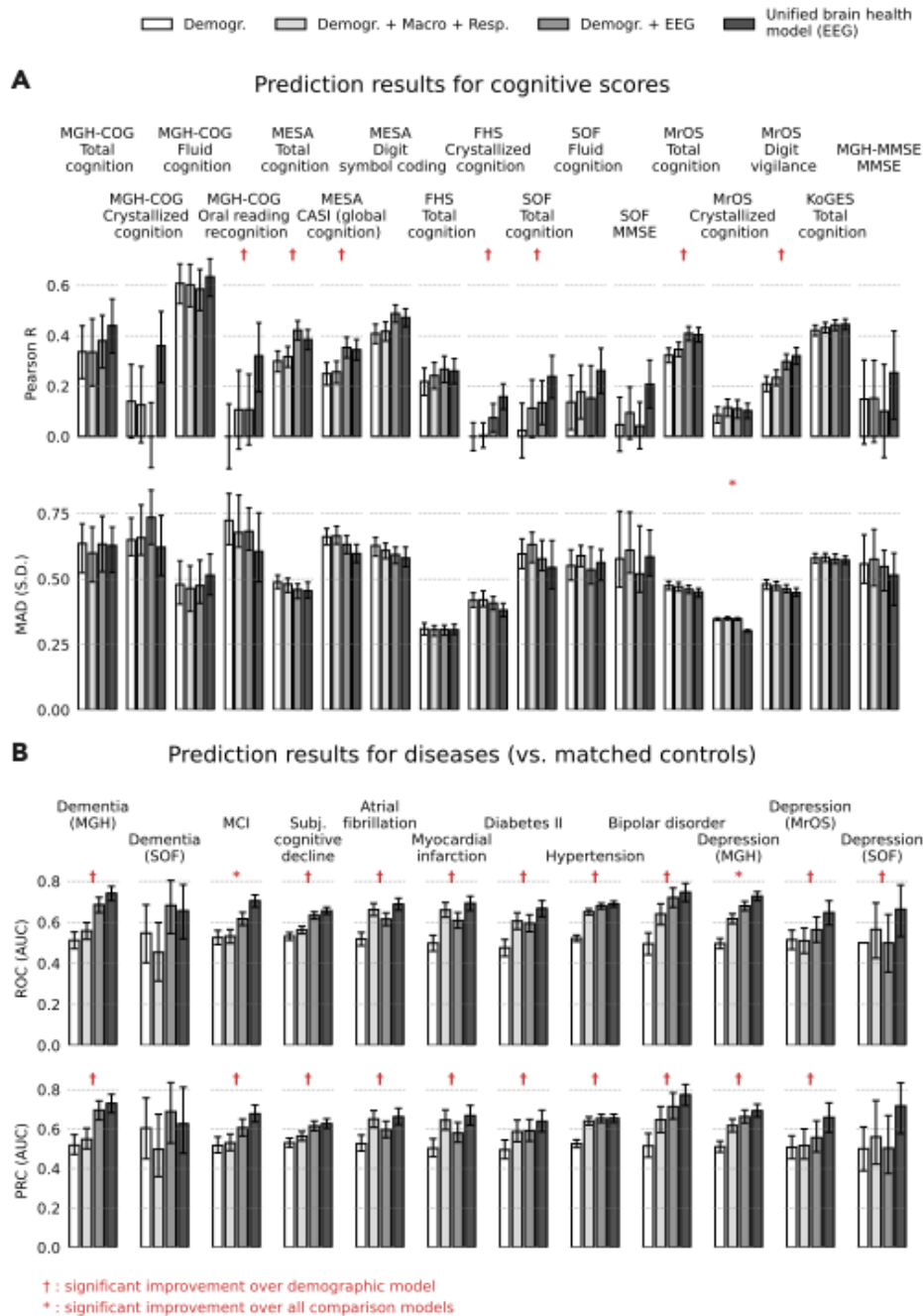
v2_1	wavelet	2	200	0.394	0.953	2E-05	0.63	0.157	0.631	0.643	0.597	0.696
v2_3	wavelet	1	199	0.393	0.471	0.0001	0.61	0.216	0.571	0.465	0.507	0.606
v2_1	wavelet	2	200	0.391	0.851	5E-05	0.63	0.139	0.643	0.539	0.561	0.641
v2_8	multitaper	1	248	0.389	0.413	0.0001	0.73	0.201	0.578	0.303	0.537	0.616
v2_2	multitaper	1	118	0.387	0.43	0.0001	0.74	0.181	0.593	0.254	0.514	0.596
v2_1	wavelet	2	200	0.383	0.978	1E-05	0.61	0.161	0.605	0.623	0.597	0.615
v2_1	wavelet	2	200	0.38	0.514	8E-06	0.59	0.183	0.577	0.45	0.416	0.572
v2_1	wavelet	2	200	0.375	0.851	2E-05	0.61	0.152	0.598	0.517	0.545	0.616
v2_8	wavelet	2	248	0.374	0.425	0.0001	0.63	0.229	0.518	0.326	0.502	0.537
v2_1	wavelet	2	200	0.37	0.726	5E-06	0.58	0.134	0.606	0.498	0.228	0.628
v2_1	wavelet	4	200	0.362	0.437	8E-06	0.46	0.162	0.563	0.422	0.354	0.526
v2_1	wavelet	4	200	0.362	0.954	5E-06	0.5	0.102	0.621	0.574	0.262	0.637
v2_1	wavelet	4	200	0.362	0.624	5E-07	0.32	0.114	0.61	0.412	0.121	0.623
v2_3	wavelet	2	199	0.36	0.407	0.0001	0.62	0.213	0.507	0.374	0.465	0.516
v2_1	wavelet	4	200	0.358	0.711	1E-06	0.38	0.112	0.604	0.467	0.198	0.597
v2_1	wavelet	2	200	0.358	0.399	0.0001	0.61	0.193	0.522	0.259	0.591	0.481
v2_1	wavelet	2	200	0.293	0.286	0.0002	0.61	0.07	0.516	0.0	0.53	0.533
v2_4	wavelet	2	248	0.292	0.274	0.0002	0.62	0.016	0.567	-0.003	0.112	0.655
v2_4	wavelet	2	248	0.284	0.255	0.0002	0.65	0.037	0.531	0.009	-0.055	0.55
v2_1	wavelet	2	200	0.273	0.288	0.0005	0.51	0.033	0.513	0.14	0.084	0.531
v2_1	multitaper	1	200	0.266	0.23	0.0001	0.75	0.016	0.516	0.109	0.435	0.536
v2_1	wavelet	2	200	0.26	0.257	0.0002	0.64	0.01	0.509	0.0	0.502	0.469

Table S1 b) Models using timeseries as input:

model_name	Model parameters (in M)	Summary performance across tasks (validation)	Summary performance (training)	Learning rate	Validation on Cohen's Kappa Sleep staging	Average correlation performance cognitive tests	Mean ROC across diseases	Mean correlation performance or sleep metrics	Performance fluid cognition in MGH	ROC dementia in MGH
v3_6LA1	67	0.413	0.594	0.0001	0.7	0.176	0.65	0.526	0.392	0.689
v3_1LA1	70	0.406	0.587	0.0001	0.71	0.215	0.598	0.358	0.622	0.624
v3_6LA2	90	0.388	0.397	0.0001	0.69	0.207	0.57	0.389	0.627	0.591
v3_0LA1	66	0.388	0.808	0.0001	0.71	0.151	0.625	0.482	0.45	0.619
v3_2A1	17	0.382	0.706	0.0001	0.69	0.159	0.606	0.431	0.595	0.618
v3_0A1	17	0.376	0.802	0.0001	0.71	0.119	0.632	0.534	0.47	0.65
v3_2LA1	48	0.37	0.7	0.0001	0.7	0.165	0.574	0.39	0.519	0.643

v3_3A1	22	0.363	0.832	0.0001	0.69	0.134	0.592	0.44	0.444	0.599
v3_1A1	22	0.362	0.447	0.0001	0.67	0.156	0.569	0.324	0.224	0.546
v3_0A3	29	0.36	0.57	0.0001	0.68	0.144	0.575	0.39	0.154	0.586
v3_5LA2	121	0.355	0.388	0.0001	0.7	0.126	0.584	0.358	0.254	0.64
v3_6A5	56	0.352	0.408	0.0001	0.68	0.128	0.577	0.324	0.403	0.6
v3_1A5	45	0.352	0.413	0.0001	0.67	0.117	0.587	0.278	0.281	0.526
v3_4A5	57	0.345	0.401	0.0001	0.69	0.115	0.575	0.334	0.231	0.624
v3_3LA1	68	0.34	0.869	0.0001	0.7	0.141	0.538	0.427	0.433	0.562
v3_3A3	33	0.334	0.414	0.0001	0.68	0.123	0.545	0.327	0.43	0.547
v3_5A5	65	0.328	0.324	0.0001	0.65	0.1	0.555	0.269	0.067	0.545
v3_3A5	45	0.323	0.414	0.0001	0.67	0.101	0.545	0.241	0.309	0.53
v3_2A3	28	0.315	0.352	0.0001	0.66	0.101	0.529	0.245	0.232	0.5
v3_0A5	41	0.314	0.476	0.0001	0.68	0.065	0.563	0.338	0.11	0.644
v3_0LA5	158	0.31	0.386	0.0001	0.67	0.045	0.576	0.231	0.034	0.606
v3_6A6	62	0.31	0.302	0.0001	0.65	0.117	0.503	0.237	0.311	0.48
v3_2LA3	94	0.292	0.25	0.0001	0.67	0.062	0.522	0.021	0.401	0.554
v3_2A5	40	0.285	0.342	0.0001	0.66	0.044	0.526	0.231	0.1	0.529
v3_1A3	34	0.284	0.282	0.0001	0.67	0.071	0.496	0.162	0.122	0.479
v3_3LA3	114	0.282	0.244	0.0001	0.68	0.038	0.526	0.0	-0.087	0.518
v3_0LA3	112	0.281	0.381	0.0001	0.68	0.03	0.532	0.309	-0.172	0.534
v3_3LA5	160	0.263	0.237	0.0001	0.7	0.041	0.485	0.055	0.162	0.49
v3_1LA3	117	0.259	0.243	0.0001	0.69	0.023	0.495	0.033	-0.004	0.467
v3_2LA5	140	0.258	0.215	0.0001	0.68	-0.021	0.537	0.01	-0.087	0.59
v3_1LA5	163	0.256	0.245	0.0001	0.69	-0.009	0.52	0.012	0.04	0.558

Extended model comparison



Supplementary Figure S5. Extended model comparison incorporating classical EEG-based machine learning: This figure compares predictive performance across four model classes: (i) demographic (age and sex) baselines, (ii) multivariate models using traditional sleep stage hypnogram and sleep-disordered breathing/respiratory summary variables, (iii) classical EEG-based machine-learning models trained on expert-defined EEG features, and (iv) our multi-task deep-learning brain health latent space using raw sleep EEG. Performance is shown for representative cognitive and disease outcomes, using correlations and median absolute error for

cognitive scores and ROC/PRC areas for disease classification. Full quantitative comparisons between classical EEG-based machine learning and deep learning are reported in Table S2

Table S2: Performance comparison between two different machine learning approaches: Deep learning (sleep EEG brain health latent space, S-BLS) vs. classic machine learning based on expert-designed features. R: Pearson correlation between observed and inferred variables; MAE: median absolute error between observed and inferred variables.

Cohort	Cognitive Measure	R (S-BLS)	R (Classic ML)	MAE (S-BLS)	MAE (Classic ML)
MGH-COG	Fluid Cognition	0.63 (0.56, 0.71)	0.59 (0.5, 0.66)	0.51 (0.43, 0.6)	0.48 (0.4, 0.57)
	Crystallized Cognition	0.36 (0.21, 0.49)*	-0.02 (-0.14, 0.12)	0.62 (0.51, 0.74)	0.74 (0.64, 0.86)
	Total Cognition	0.44 (0.33, 0.55)	0.38 (0.27, 0.48)	0.63 (0.53, 0.7)	0.63 (0.53, 0.74)
	Picture Sequence Memory	0.53 (0.43, 0.63)	0.51 (0.4, 0.6)	0.6 (0.53, 0.69)	0.61 (0.54, 0.71)
	Pattern Comp. Process. Speed	0.6 (0.5, 0.68)	0.63 (0.54, 0.71)	0.53 (0.44, 0.61)	0.54 (0.42, 0.65)
	Dimensional Change Card Sort	0.35 (0.22, 0.48)	0.3 (0.18, 0.43)	0.6 (0.54, 0.68)	0.64 (0.55, 0.71)
	Oral Reading Recognition	0.32 (0.18, 0.46)	0.11 (-0.03, 0.25)	0.61 (0.49, 0.75)	0.68 (0.61, 0.77)
MESA	Total Cognition	0.39 (0.35, 0.42)	0.42 (0.38, 0.46)	0.46 (0.43, 0.49)	0.46 (0.43, 0.48)
	Digit Symbol Coding	0.47 (0.44, 0.51)	0.49 (0.45, 0.52)	0.58 (0.55, 0.62)	0.59 (0.56, 0.62)
	CASI (global cognition)	0.34 (0.3, 0.39)	0.35 (0.31, 0.39)	0.6 (0.56, 0.63)	0.63 (0.6, 0.67)
	Digit Span F	0.18 (0.14, 0.23)	0.2 (0.15, 0.24)	0.71 (0.67, 0.75)	0.7 (0.66, 0.73)
	Digit Span B	0.21 (0.17, 0.26)	0.17 (0.13, 0.22)	0.67 (0.62, 0.7)	0.66 (0.62, 0.69)
FHS	Fluid Cognition	0.31 (0.26, 0.36)	0.33 (0.29, 0.38)	0.37 (0.34, 0.39)	0.35 (0.34, 0.38)
	Crystallized Cognition	0.16 (0.11, 0.21)	0.07 (0.02, 0.13)	0.38 (0.36, 0.41)	0.41 (0.38, 0.43)
	Total Cognition	0.26 (0.21, 0.31)	0.27 (0.21, 0.32)	0.31 (0.29, 0.33)	0.31 (0.29, 0.32)
	Logical Mem - DR	0.23 (0.18, 0.28)	0.2 (0.15, 0.26)	0.63 (0.59, 0.69)	0.65 (0.6, 0.69)

Cohort	Cognitive Measure	R (S-BLS)	R (Classic ML)	MAE (S-BLS)	MAE (Classic ML)
SOF	MMSE - Orientation (Time Rec)	-0.01 (-0.05, 0.05)	0.01 (-0.04, 0.06)	0.05 (0.05, 0.05)*	0.12 (0.12, 0.12)
	Trail Making A	0.28 (0.21, 0.34)	0.3 (0.25, 0.35)	0.47 (0.45, 0.5)	0.47 (0.44, 0.49)
	Fluid Cognition	0.26 (0.17, 0.35)	0.15 (0.01, 0.28)	0.56 (0.5, 0.61)	0.54 (0.48, 0.62)
	Crystallized Cognition	0.09 (0.0, 0.18)	-0.05 (-0.13, 0.02)	0.6 (0.55, 0.65)	0.72 (0.65, 0.78)
	Total Cognition	0.24 (0.15, 0.32)	0.14 (0.05, 0.22)	0.54 (0.46, 0.64)	0.58 (0.53, 0.65)
MrOS	Trails B	0.2 (0.11, 0.3)	0.13 (0.03, 0.23)	0.51 (0.46, 0.57)	0.51 (0.46, 0.55)
	Short MMSE	0.09 (0.0, 0.18)*	-0.09 (-0.16, -0.02)	0.6 (0.55, 0.65)*	0.77 (0.66, 0.78)
	MMSE total	0.21 (0.11, 0.3)	0.04 (-0.05, 0.13)	0.58 (0.51, 0.69)	0.52 (0.45, 0.7)
	Fluid Cognition	0.36 (0.33, 0.39)	0.37 (0.34, 0.4)	0.5 (0.48, 0.51)	0.5 (0.48, 0.51)
	Crystallized Cognition	0.1 (0.07, 0.13)	0.11 (0.07, 0.15)	0.3 (0.3, 0.31)*	0.35 (0.34, 0.35)
	Total Cognition	0.4 (0.38, 0.43)	0.41 (0.38, 0.44)	0.45 (0.43, 0.46)	0.46 (0.45, 0.48)
	Trails B	0.35 (0.32, 0.38)	0.35 (0.32, 0.38)	0.52 (0.5, 0.54)	0.54 (0.52, 0.56)
	Teng - Naming	-0.0 (-0.04, 0.03)	-0.01 (-0.04, 0.02)	0.11 (0.11, 0.11)*	0.16 (0.16, 0.16)
	Digit Vigilance score	0.32 (0.29, 0.35)	0.3 (0.27, 0.33)	0.45 (0.43, 0.47)	0.46 (0.45, 0.48)
	Teng 3MS	0.27 (0.24, 0.3)	0.28 (0.25, 0.32)	0.49 (0.48, 0.51)	0.51 (0.49, 0.53)
KoGES	Fluid Cognition	0.46 (0.44, 0.48)	0.45 (0.43, 0.47)	0.57 (0.56, 0.59)	0.58 (0.56, 0.6)
	Crystallized Cognition	0.22 (0.2, 0.25)	0.22 (0.2, 0.25)	0.6 (0.59, 0.62)	0.61 (0.6, 0.63)
	Total Cognition	0.45 (0.42, 0.47)	0.44 (0.42, 0.46)	0.57 (0.56, 0.59)	0.58 (0.56, 0.6)
	Trails A	0.48 (0.46, 0.5)	0.48 (0.46, 0.5)	0.4 (0.39, 0.41)	0.41 (0.4, 0.42)
	Visual Reproductions - DR	0.33 (0.31, 0.36)	0.32 (0.3, 0.35)	0.69 (0.67, 0.71)	0.69 (0.67, 0.71)

Cohort	Cognitive Measure	R (S-BLS)	R (Classic ML)	MAE (S-BLS)	MAE (Classic ML)
	Digit Symbol - Coding	0.52 (0.5, 0.54)	0.53 (0.51, 0.54)	0.58 (0.56, 0.6)	0.57 (0.55, 0.59)
	Stroop - Word Reading	0.34 (0.27, 0.4)	0.35 (0.28, 0.41)	0.55 (0.54, 0.57)	0.54 (0.52, 0.56)
MGH-MMSE	MMSE total	0.25 (0.07, 0.42)	0.1 (-0.08, 0.28)	0.51 (0.4, 0.6)	0.55 (0.46, 0.61)

Part B: Diseases; ROC: receiver operating curve area under the curve; PRC: precision-recall curve area under the curve.

Cohort	Disease	ROC (S-BLS)	ROC (Classic ML)	PRC (S-BLS)	PRC (Classic ML)
MGH	Atrial fibrillation	0.69 (0.66, 0.72)*	0.62 (0.58, 0.65)	0.66 (0.62, 0.71)	0.6 (0.56, 0.64)
	Bipolar disorder	0.75 (0.7, 0.79)	0.72 (0.67, 0.77)	0.77 (0.72, 0.83)	0.71 (0.65, 0.78)
	Dementia	0.74 (0.71, 0.77)	0.69 (0.65, 0.72)	0.73 (0.68, 0.78)	0.7 (0.65, 0.74)
	Depression	0.73 (0.71, 0.75)*	0.68 (0.66, 0.7)	0.69 (0.66, 0.73)	0.66 (0.63, 0.69)
	Diabetes II	0.67 (0.63, 0.71)	0.6 (0.55, 0.64)	0.64 (0.59, 0.7)	0.59 (0.54, 0.65)
	Hypertension	0.69 (0.68, 0.7)	0.68 (0.66, 0.69)	0.65 (0.63, 0.68)	0.65 (0.63, 0.67)
	MCI	0.7 (0.67, 0.73)*	0.62 (0.58, 0.65)	0.68 (0.64, 0.72)	0.61 (0.57, 0.65)
	Myocardial infarction	0.69 (0.66, 0.73)*	0.61 (0.57, 0.65)	0.67 (0.62, 0.72)	0.58 (0.53, 0.63)
	Subj. cognitive decline	0.66 (0.64, 0.67)	0.63 (0.62, 0.65)	0.63 (0.6, 0.65)	0.62 (0.59, 0.64)
MrOS	Depression	0.65 (0.59, 0.7)	0.56 (0.5, 0.63)	0.66 (0.58, 0.73)	0.56 (0.49, 0.64)
SOF	Dementia	0.66 (0.52, 0.79)	0.68 (0.55, 0.81)	0.63 (0.48, 0.81)	0.69 (0.53, 0.84)
	Depression	0.66 (0.53, 0.78)	0.5 (0.36, 0.64)	0.72 (0.58, 0.84)	0.5 (0.37, 0.67)

Survival analysis

Methodology

Survival analyses were conducted separately in four cohorts (MrOS, FHS, MGH, BIDMC). For each participant, the outcome was all-cause mortality, with time-to-event defined as days from the sleep study to death or censoring. All analyses used the cohort-provided follow-up durations and event indicators.

Preprocessing

- Participants missing follow-up time, event status, or age were excluded.
- Age was standardized within-cohort to yield an age z-score (age_z), used in all age-adjusted models.
- Candidate predictors included the brain health score (BHS), Brain Age Index (BAI), REM sleep percentage, spindle density, and total cognition observed from the neuropsychological test batteries, depending on cohort availability.
- All predictors were z-scored.

Cox Proportional Hazards Models

We used the Cox proportional hazards model (lifelines CoxPHFitter) to test associations between each predictor and mortality.

All models were adjusted for age_z:

$$h(t) = h_0(t) \exp(\beta_{\text{age}} \text{age}_z + \beta_{\text{feature}} X)$$

For each feature we obtained:

- hazard ratio per SD increase
- 95% Wald confidence interval
- p-value
- Harrell's C-index with bootstrap 95% CI (10,000 bootstrap replicates).

Model Comparison and Added Value

To quantify added prognostic value over age, we compared age-only models with age + feature models using:

- Likelihood ratio test
- ΔAIC (Akaike Information Criterion)
- ΔBIC (Bayesian Information Criterion)

Large positive $\Delta\text{AIC}/\Delta\text{BIC}$ values indicate improved fit.

Time-dependent discrimination was assessed using IPCW C(t) at pre-specified horizons (5, 8, 10, 12, and 15 years depending on cohort).

Absolute Risk Estimation

For interpretability, we computed model-based absolute mortality risk at 5-year horizons for representative ages (e.g., 65, 75, 70, 80 years) using:

$$\hat{S}(t | X) = S_0(t)^{\exp(\beta^T X)}$$

Quartiles of the brain health score were used to illustrate age-adjusted differences in predicted risk.

Reclassification Analyses (NRI and IDI)

To assess individual-level improvements in risk predictions when adding BHS to age:

- Net Reclassification Improvement (NRI) and
- Integrated Discrimination Improvement (IDI)

were computed using category-free definitions at 5, 8, 10, and 12 years.

Events were defined as deaths occurring before the horizon. Non-events survived beyond the horizon. Analyses were restricted to participants with usable predictions for both models.

Testing whether EEG and cognition features differ from BHS

To formally compare the mortality associations of BHS versus REM%, spindle density, or total cognition, we fit joint Cox models containing age_z, BHS, and the comparator feature.

Equality of coefficients was tested using a Wald test on $\beta_{\text{feature}} - \beta_{\text{BHS}}$.

P-values are reported.

Analyses were implemented in Python 3.12, using lifelines for Cox models and IPCW, numpy/pandas for data handling, and custom utilities for bootstrap confidence intervals and reclassification metrics.

MrOS survival analysis (n = 2,454)

Participants (age 67–96 years, mean 76.3) were followed for a median of 12.15 years, with 50% mortality by the median follow-up. In age-adjusted Cox models, the brain health score (BHS) had a hazard ratio (HR) of **0.69 per SD** ($p < 0.0001$), comparable to total cognition (HR 0.70) and lower than REM% and spindle density (both HR 0.90). The EEG-based Brain Age Index (BAI) showed the opposite-direction association (HR **1.09**, $p = 0.0007$). Discrimination improved from an age-only C-index of **0.674** to **0.691** with BHS and **0.676** with BAI; combining age + BHS + BAI yielded **0.692**. Likelihood-ratio tests showed a larger improvement for adding BHS to age (LR = **138.0**, $p = 7.2 \times 10^{-32}$) than for adding BAI (LR = **11.6**, $p = 6.5 \times 10^{-4}$). Time-dependent discrimination at 5, 10, and 15 years increased from **0.688/0.721/0.774** (age only) to **0.715/0.740/0.798** when adding BHS. Model-based 5-year absolute mortality risk at age 70 ranged from **0.107** (lowest BHS quartile) to **0.045** (highest), consistent with Kaplan–Meier estimates (0.264 to 0.047 across quartiles). Reclassification analyses showed net reclassification improvement (NRI) values of **41.2%**, **35.8%**, **32.7%**, and **32.5%** at 5, 8, 10, and 12 years, with corresponding IDI values of **0.0185**, **0.0248**, **0.0273**, and **0.0285**.

FHS survival analysis (n = 744)

Participants (age 39–81 years, mean 57.7) were followed for a median of 11.90 years, with 9.1% mortality by that point. In age-adjusted Cox models, BHS had an HR of **0.85 per SD** ($p = 0.28$). Other EEG metrics showed HRs near 1.0, and BAI had an HR of **1.01** ($p = 0.97$). Discrimination

for age alone was **0.753** and was **0.755** with BHS and **0.754** with BAI. Likelihood-ratio comparisons showed minimal differences when adding either predictor to age (LR = **1.17** for BHS; LR = **0.00** for BAI). Time-dependent discrimination at 5 and 10 years was **0.678/0.789** for age alone and **0.680/0.789** with BHS. Model-based 5-year absolute mortality risk at age 65 ranged from **0.022** (lowest BHS quartile) to **0.014** (highest), and at age 75 from **0.057** to **0.038**. Reclassification metrics at 5, 8, 10, and 12 years yielded NRI values of **13.1%**, **22.1%**, **0.3%**, and **-5.9%**, with corresponding IDI values of **-0.0003**, **0.0024**, **0.0014**, and **0.0012**.

MGH survival analysis (n = 8,285)

Participants (age 18–90 years, mean 50.9) were followed for a median of 5.37 years, with 3.6% mortality by that point. In age-adjusted Cox models, BHS had an HR of **0.65 per SD** ($p < 0.001$). Other EEG metrics showed HRs between **0.63** and **0.75**, and BAI showed an HR of **1.50** ($p < 0.001$). Discrimination for age alone was **0.785**, increasing to **0.806** with BHS and **0.803** with BAI. Likelihood-ratio tests comparing models with and without each predictor yielded LR values of **54.28** for BHS and **40.50** for BAI when added to age. At 5 and 10 years, time-dependent discrimination increased from **0.785/0.817** (age alone) to **0.807/0.835** with BHS. Model-estimated 5-year absolute mortality risk at age 65 ranged from **0.063** (lowest BHS quartile) to **0.022** (highest), and at age 75 from **0.123** to **0.044**. Reclassification analyses showed NRI values of **22.3%**, **19.9%**, **18.4%**, and **16.3%** at 5, 8, 10, and 12 years, with corresponding IDI values of **0.0093**, **0.0119**, **0.0135**, and **0.0123**.

BIDMC survival analysis (n = 6,483)

Participants (age 50–98 years, mean 66.0) were followed for a median of 2.96 years, with 9.7% mortality by that point. In age-adjusted Cox models, the BHS had an HR of 0.76 per SD ($p < 0.001$). Other EEG metrics showed HRs ranging from 0.80 to 0.91, and BAI showed an HR of 1.08 ($p = 0.049$). Discrimination for age alone was 0.608, increasing to 0.617 with BHS and 0.612 with BAI. Likelihood-ratio tests showed LR = 17.71 for adding BHS to age and LR = 3.94 for adding BAI to age. Time-dependent discrimination at 5 years increased from 0.622 (age alone) to 0.646 with BHS. Model-estimated 5-year mortality risk at age 65 ranged from 0.238 (lowest BHS quartile) to 0.125 (highest), and at age 75 from 0.279 to 0.148. Reclassification analysis at 5 years yielded an NRI of 29.9% and an IDI of 0.0098.

Comparison of hazard ratios between brain health score and conventional EEG or cognitive metrics

To assess whether the brain health score (BHS) differed significantly in prognostic strength from conventional metrics, we compared age-adjusted hazard ratios within each cohort. Across cohorts, the p-values for testing equality of HRs (conventional metric vs. BHS) were as follows:

	MrOS	FHS	MGH	BIDMC
REM sleep percentage	1.6×10^{-13}	0.58	0.040	0.017
Fast spindle density	4.5×10^{-13}	0.34	0.997	0.934
Neuropsychological total cognitive score	0.56	0.53	Not available	Not available

These tests provide a cohort-specific statistical comparison of each metric's association with survival relative to BHS.

Comparison Between Sleep Brain Health Latent Space and EEG-Based Brain Age Index

Because BAI was originally developed on MGH data and brain health latent space (S-BLS) was trained using EEG from multiple cohorts, cross-sectional cognition and disease comparisons do not constitute fully matched external evaluations. Nevertheless, these analyses are informative for understanding the relative behavior and potential complementarities of the two biomarkers across diverse cognitive domains and clinical conditions.

Therefore, we present complete S-BLS –BAI comparison table for cognition and disease prediction (Supplementary Table S3).

For a fully external dataset comparison, we conducted survival analysis in the BIDMC cohort, the only dataset not used in training either biomarker. Results, including likelihood ratio tests, AIC/BIC, C-index, and time-dependent discrimination, are detailed in Supplementary Table S4.

For each subject, we computed S-BLS and BAI values and fit age-adjusted Cox proportional hazards models with identical covariate structure:

- Model 1: age
- Model 2: age + BAI
- Model 3: age + S-BLS
- Model 4: age + BAI + S-BLS

Model fit was compared via:

- Likelihood ratio tests
- Δ AIC / Δ BIC
- Δ -2LL
- Harrell's C-index with bootstrapped 95% CI
- Integrated time-dependent discrimination C(t)

Table S3: Performance comparison between two different machine learning approaches: Deep learning (brain health latent space, S-BLS) vs. brain age index.

R: Pearson correlation between observed and inferred variables; MAE: median absolute error between observed and inferred variables.

Cohort	Cognitive Measure	R (S-BLS)	R (Demographics + BAI)	MAE (S-BLS)	MAE (Demographics + BAI)
MGH-COG	Fluid Cognition	0.63 (0.56, 0.71)	0.61 (0.54, 0.69)	0.51 (0.44, 0.6)	0.48 (0.43, 0.59)
	Crystallized Cognition	0.36 (0.21, 0.5)	0.13 (-0.02, 0.27)	0.62 (0.51, 0.74)	0.66 (0.53, 0.78)
	Total Cognition	0.44 (0.33, 0.55)	0.35 (0.24, 0.45)	0.63 (0.53, 0.7)	0.61 (0.55, 0.71)

Cohort	Cognitive Measure	R (S-BLS)	R (Demographics + BAI)	MAE (S-BLS)	MAE (Demographics + BAI)
MESA	Picture Sequence Memory	0.53 (0.43, 0.63)	0.45 (0.34, 0.55)	0.6 (0.53, 0.69)	0.69 (0.59, 0.77)
	Pattern Comp. Process. Speed	0.6 (0.5, 0.68)	0.61 (0.52, 0.69)	0.53 (0.43, 0.61)	0.5 (0.43, 0.6)
	Dimensional Change Card Sort	0.35 (0.23, 0.48)	0.36 (0.24, 0.48)	0.6 (0.52, 0.68)	0.59 (0.51, 0.69)
	Oral Reading Recognition	0.32 (0.18, 0.45)*	-0.13 (-0.25, 0.0)	0.61 (0.49, 0.75)	0.72 (0.63, 0.86)
	Total Cognition	0.39 (0.35, 0.42)*	0.3 (0.26, 0.34)	0.46 (0.43, 0.49)	0.48 (0.46, 0.52)
	Digit Symbol Coding	0.47 (0.44, 0.51)	0.41 (0.37, 0.45)	0.58 (0.55, 0.62)	0.63 (0.59, 0.66)
	CASI (global cognition)	0.34 (0.3, 0.39)*	0.25 (0.2, 0.29)	0.6 (0.56, 0.63)	0.66 (0.63, 0.69)
FHS	Digit Span F	0.18 (0.14, 0.23)	0.13 (0.09, 0.18)	0.71 (0.67, 0.75)	0.69 (0.65, 0.73)
	Digit Span B	0.21 (0.17, 0.26)	0.13 (0.09, 0.18)	0.67 (0.62, 0.7)	0.67 (0.63, 0.7)
	Fluid Cognition	0.31 (0.26, 0.36)	0.31 (0.26, 0.36)	0.37 (0.34, 0.39)	0.36 (0.34, 0.38)
	Crystallized Cognition	0.16 (0.11, 0.21)*	-0.05 (-0.1, 0.0)	0.38 (0.36, 0.41)	0.42 (0.39, 0.46)
	Total Cognition	0.26 (0.21, 0.31)	0.23 (0.17, 0.28)	0.31 (0.29, 0.33)	0.31 (0.29, 0.33)
	Logical Mem - DR	0.23 (0.18, 0.28)	0.16 (0.11, 0.21)	0.63 (0.59, 0.69)	0.63 (0.59, 0.68)
	MMSE - Orientation (Time Rec)	-0.01 (-0.05, 0.05)	-0.09 (-0.14, -0.04)	0.05 (0.05, 0.05)*	0.14 (0.14, 0.15)
SOF	Trail Making A	0.28 (0.21, 0.34)	0.27 (0.21, 0.32)	0.47 (0.45, 0.5)	0.48 (0.46, 0.51)
	Fluid Cognition	0.26 (0.17, 0.35)	0.11 (0.01, 0.22)	0.56 (0.5, 0.61)	0.56 (0.5, 0.63)
	Crystallized Cognition	0.09 (0.01, 0.18)*	-0.16 (-0.25, -0.06)	0.6 (0.55, 0.65)	0.79 (0.64, 0.79)
	Total Cognition	0.24 (0.15, 0.32)*	-0.02 (-0.13, 0.09)	0.54 (0.46, 0.65)	0.6 (0.55, 0.67)
	Trails B	0.2 (0.11, 0.3)	0.12 (0.03, 0.21)	0.51 (0.46, 0.57)	0.52 (0.46, 0.58)
	Short MMSE	0.09 (0.0, 0.18)*	-0.16 (-0.25, -0.06)	0.6 (0.55, 0.65)	0.79 (0.65, 0.79)
	MMSE total	0.21 (0.11, 0.3)*	-0.07 (-0.15, 0.01)	0.58 (0.51, 0.69)	0.58 (0.43, 0.77)
MrOS	Fluid Cognition	0.36 (0.33, 0.39)	0.31 (0.28, 0.34)	0.5 (0.48, 0.51)	0.51 (0.5, 0.53)
	Crystallized Cognition	0.1 (0.07, 0.13)	0.08 (0.05, 0.11)	0.3 (0.3, 0.31)*	0.35 (0.34, 0.35)

Cohort	Cognitive Measure	R (S-BLS)	R (Demographics + BAI)	MAE (S-BLS)	MAE (Demographics + BAI)
	Total Cognition	0.4 (0.38, 0.43)*	0.33 (0.3, 0.36)	0.45 (0.43, 0.46)	0.48 (0.45, 0.5)
	Trails B	0.35 (0.32, 0.38)	0.29 (0.27, 0.32)	0.52 (0.5, 0.54)	0.55 (0.53, 0.57)
	Teng - Naming	-0.0 (-0.04, 0.03)	-0.04 (-0.07, -0.02)	0.11 (0.11, 0.11)*	0.16 (0.16, 0.16)
	Digit Vigilance score	0.32 (0.29, 0.35)*	0.21 (0.18, 0.24)	0.45 (0.43, 0.47)	0.48 (0.46, 0.5)
	Teng 3MS	0.27 (0.24, 0.3)	0.24 (0.21, 0.27)	0.49 (0.48, 0.51)	0.52 (0.51, 0.54)
KoGES	Fluid Cognition	0.46 (0.44, 0.48)	0.43 (0.41, 0.45)	0.57 (0.56, 0.59)	0.59 (0.58, 0.61)
	Crystallized Cognition	0.22 (0.2, 0.24)	0.21 (0.19, 0.24)	0.6 (0.59, 0.62)	0.62 (0.6, 0.64)
	Total Cognition	0.45 (0.42, 0.47)	0.42 (0.4, 0.44)	0.57 (0.56, 0.59)	0.58 (0.56, 0.6)
	Trails A	0.48 (0.46, 0.5)	0.46 (0.44, 0.48)	0.4 (0.39, 0.41)	0.41 (0.4, 0.43)
	Visual Reproductions - DR	0.33 (0.31, 0.36)	0.32 (0.29, 0.34)	0.69 (0.67, 0.71)	0.7 (0.67, 0.72)
	Digit Symbol - Coding	0.52 (0.5, 0.54)	0.51 (0.49, 0.53)	0.58 (0.56, 0.6)	0.58 (0.56, 0.59)
	Stroop - Word Reading	0.34 (0.27, 0.4)	0.34 (0.28, 0.4)	0.55 (0.54, 0.57)	0.55 (0.53, 0.56)
MGH-MMSE	MMSE total	0.25 (0.07, 0.42)	0.17 (-0.0, 0.33)	0.51 (0.4, 0.6)	0.55 (0.46, 0.64)

Cohort	Disease	ROC (Philosopher)	ROC (Demographics + BAI)	PRC (Philosopher)	PRC (Demographics + BAI)
MGH	Atrial fibrillation	0.69 (0.66, 0.72)*	0.52 (0.49, 0.55)	0.66 (0.62, 0.71)*	0.52 (0.48, 0.56)
	Bipolar disorder	0.75 (0.7, 0.79)*	0.62 (0.57, 0.67)	0.77 (0.72, 0.83)*	0.64 (0.58, 0.71)
	Dementia	0.74 (0.71, 0.78)*	0.58 (0.54, 0.62)	0.73 (0.68, 0.78)*	0.57 (0.52, 0.62)
	Depression	0.73 (0.71, 0.75)*	0.54 (0.52, 0.57)	0.69 (0.66, 0.73)*	0.54 (0.51, 0.57)
	Diabetes II	0.67 (0.63, 0.71)*	0.49 (0.44, 0.53)	0.64 (0.59, 0.7)*	0.5 (0.45, 0.55)
	Hypertension	0.69 (0.68, 0.7)*	0.52 (0.5, 0.53)	0.65 (0.63, 0.68)*	0.52 (0.5, 0.54)
	MCI	0.7 (0.67, 0.73)*	0.54 (0.5, 0.57)	0.68 (0.64, 0.72)*	0.54 (0.5, 0.58)
	Myocardial infarction	0.69 (0.66, 0.73)*	0.49 (0.45, 0.53)	0.67 (0.62, 0.72)*	0.5 (0.46, 0.55)
	Subj. cognitive decline	0.66 (0.64, 0.67)*	0.56 (0.54, 0.58)	0.63 (0.6, 0.65)*	0.55 (0.53, 0.58)
MrOS	Depression	0.65 (0.59, 0.7)	0.58 (0.52, 0.64)	0.66 (0.58, 0.73)	0.6 (0.52, 0.68)

Cohort	Disease	ROC (Philosopher)	ROC (Demographics + BAI)	PRC (Philosopher)	PRC (Demographics + BAI)
SOF	Dementia	0.66 (0.52, 0.78)	0.6 (0.46, 0.74)	0.63 (0.48, 0.81)	0.58 (0.43, 0.76)
	Depression	0.66 (0.53, 0.78)	0.41 (0.28, 0.54)	0.72 (0.58, 0.83)	0.47 (0.34, 0.63)

Results survival analysis BHS vs. BAI

MrOS (N = 2,454)

Both BHS (HR = 0.69) and BAI (HR = 1.09) were significant predictors of mortality.

Model-comparison tests showed that each biomarker added information when entered into a model already containing age and the other biomarker:

- Adding BHS to age+BAI improved fit (LR $p = 2.3 \times 10^{-30}$).
- Adding BAI to age+BHS improved fit ($p = 0.030$).

This pattern suggests partially overlapping but non-identical predictive information.

Discrimination was highest for models including BHS (C-index ≈ 0.69), with the combined model showing the best overall performance.

FHS (N = 744)

In FHS, neither BHS nor BAI reached statistical significance in age-adjusted Cox models. This result likely is due to the characteristics of the cohort: FHS had a very low number of deaths (10 events at 5 years; 22 at 8 years; 41 at 10 years), markedly lower than the other cohorts. Such sparse event counts dramatically limit statistical power to detect hazard ratios near the effect sizes observed in the other datasets. FHS participants were also substantially younger and generally healthier at baseline, yielding a narrower distribution of mortality risk and attenuating EEG–mortality associations. Notably, despite non-significant hazard ratios, the C-index remained similar to the other cohorts (≈ 0.75), indicating that BHS still preserved meaningful rank-order discrimination even in a low-event, low-risk population.

MGH (N = 8,285; cohort used in training both models, BHS cross-validated here)

Both biomarkers were significant predictors (BHS HR = 0.65; BAI HR = 1.50).

As in MrOS, each biomarker improved model fit when added to a model already containing age and the other biomarker, indicating complementary signal:

- BHS added to age+BAI: LR $p = 7.5 \times 10^{-11}$
- BAI added to age+BHS: LR $p = 8.8 \times 10^{-8}$

The combined model achieved the highest discrimination (C-index = 0.818).

BIDMC (N = 6,483; fully external to both models).

Both BHS (HR = 0.76) and BAI (HR = 1.08) were significant predictors in age-adjusted Cox models.

- Adding **BHS** to age+BAI improved fit (LR $p = 1.5 \times 10^{-4}$).
- Adding **BAI** to age+BHS did **not** improve fit ($p = 0.42$).

BHS achieved the highest discrimination (C-index = 0.617; 5-year C(t) = 0.646).

Threshold-based performance analysis

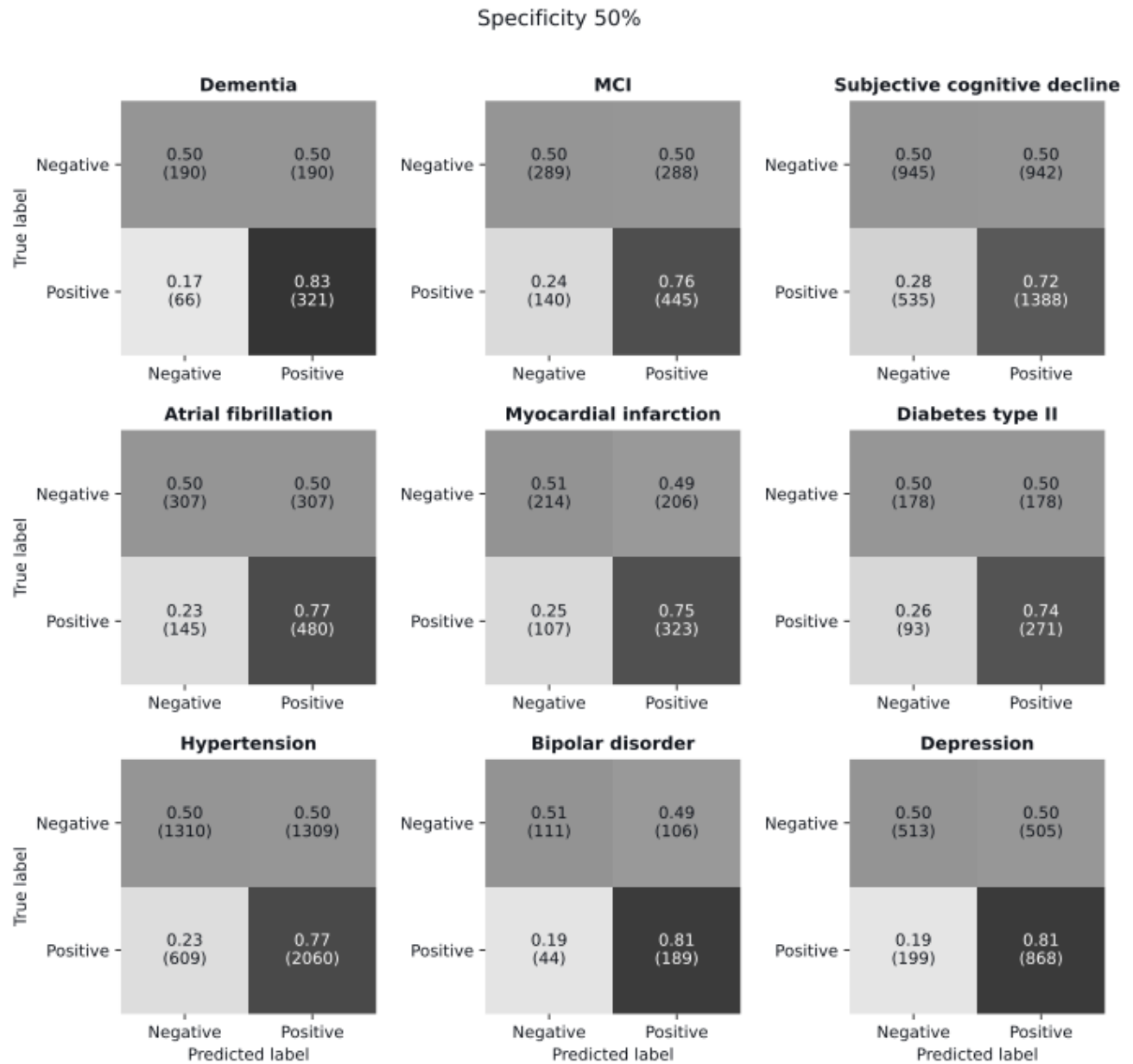


Figure S6. Confusion matrices at specificity 50%

Table S4. Specificity 50% summary table

	Atrial fibrillation	Bipolar disorder	Dementia	Depression	Diabetes type II	Hypertension	MCI	Myocardial infarction	Subjective cognitive decline
Number of samples	1239	450	767	2085	720	5288	1162	850	3810
N positives	625	233	387	1067	364	2669	585	430	1923
N negatives	614	217	380	1018	356	2619	577	420	1887
Prevalence	0.5	0.52	0.5	0.51	0.51	0.5	0.5	0.51	0.5
Threshold	0.48	0.41	0.44	0.43	0.49	0.47	0.47	0.49	0.49
ROC AUC (95% CI)	0.69 (0.66- 0.70)	0.75 (0.73- 0.75)	0.74 (0.73- 0.75)	0.73 (0.71- 0.74)	0.67 (0.63- 0.67)	0.69 (0.69- 0.70)	0.70 (0.69- 0.72)	0.69 (0.68- 0.73)	0.66 (0.64-0.66)
PR AUC (95% CI)	0.66 (0.62- 0.70)	0.77 (0.74- 0.80)	0.73 (0.71- 0.72)	0.69 (0.69- 0.71)	0.64 (0.60- 0.66)	0.65 (0.66- 0.67)	0.68 (0.66- 0.69)	0.67 (0.64- 0.70)	0.63 (0.61-0.64)
Positive Predictive Value	0.61	0.64	0.63	0.63	0.6	0.61	0.61	0.61	0.6
Negative Predictive Value	0.68	0.72	0.74	0.72	0.66	0.68	0.67	0.67	0.64
Predicted Positive Rate	0.64	0.66	0.67	0.66	0.62	0.64	0.63	0.62	0.61

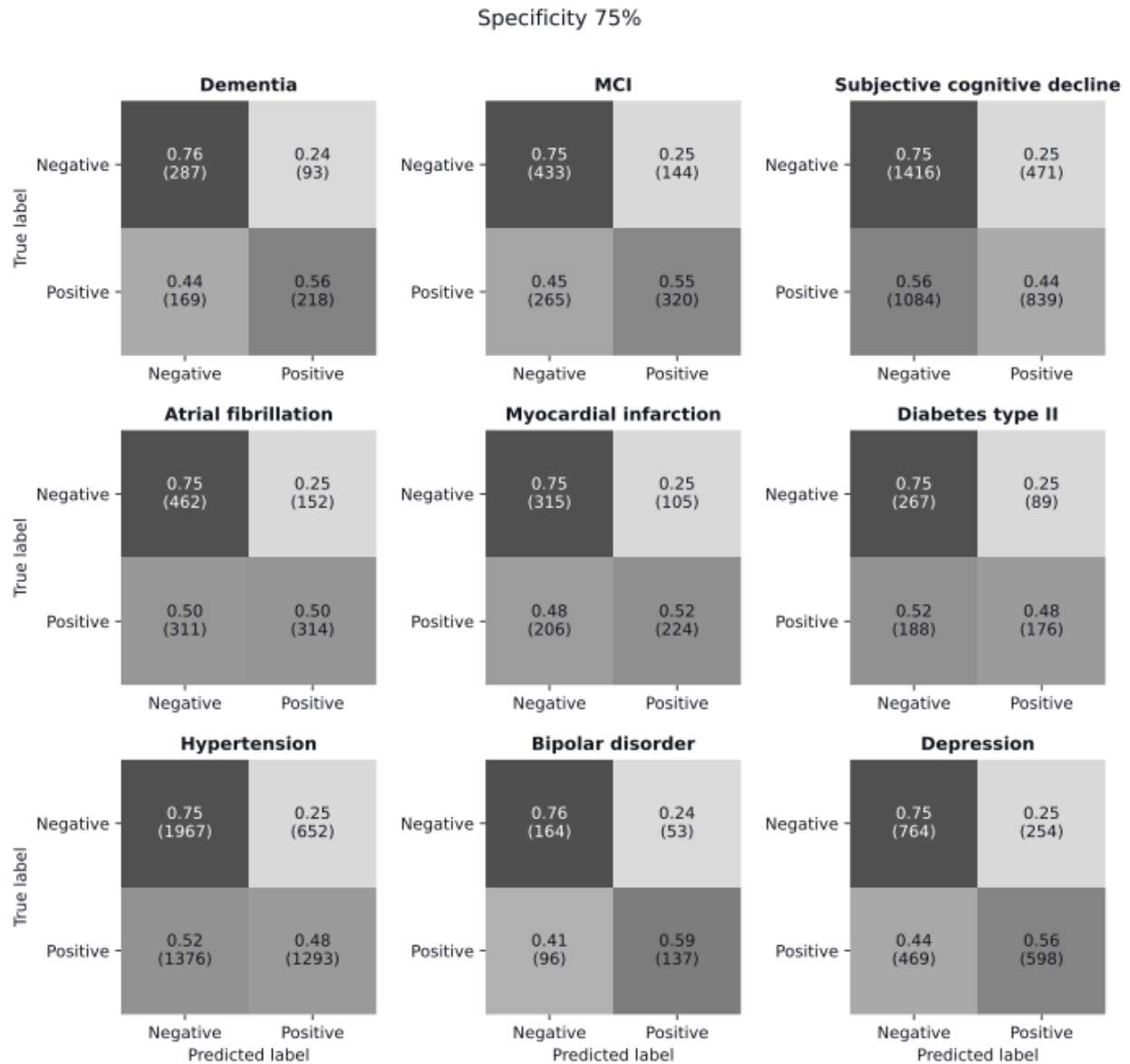


Figure S7. Confusion matrices at specificity 75%

Table S5. Specificity 0.75 summary table

	Atrial fibrillation	Bipolar disorder	Dementia	Depression	Diabetes type II	Hypertension	MCI	Myocardial infarction	Subjective cognitive decline
Number of samples	1239	450	767	2085	720	5288	1162	850	3810
N positives	625	233	387	1067	364	2669	585	430	1923
N negatives	614	217	380	1018	356	2619	577	420	1887
Prevalence	0.5	0.52	0.5	0.51	0.51	0.5	0.5	0.51	0.5
Threshold	0.62	0.59	0.61	0.64	0.61	0.64	0.61	0.62	0.64
ROC AUC (95% CI)	0.69 (0.66- 0.70)	0.75 (0.73- 0.75)	0.74 (0.73- 0.75)	0.73 (0.71- 0.74)	0.67 (0.63- 0.67)	0.69 (0.69- 0.70)	0.70 (0.69- 0.72)	0.69 (0.68- 0.73)	0.66 (0.64-0.66)
PR AUC (95% CI)	0.66 (0.62- 0.70)	0.77 (0.74- 0.80)	0.73 (0.71- 0.72)	0.69 (0.69- 0.71)	0.64 (0.60- 0.66)	0.65 (0.66- 0.67)	0.68 (0.66- 0.69)	0.67 (0.64- 0.70)	0.63 (0.61-0.64)
Positive Predictive Value	0.67	0.72	0.7	0.7	0.66	0.66	0.69	0.68	0.64
Negative Predictive Value	0.6	0.63	0.63	0.62	0.59	0.59	0.62	0.6	0.57
Predicted Positive Rate	0.38	0.42	0.41	0.41	0.37	0.37	0.4	0.39	0.34

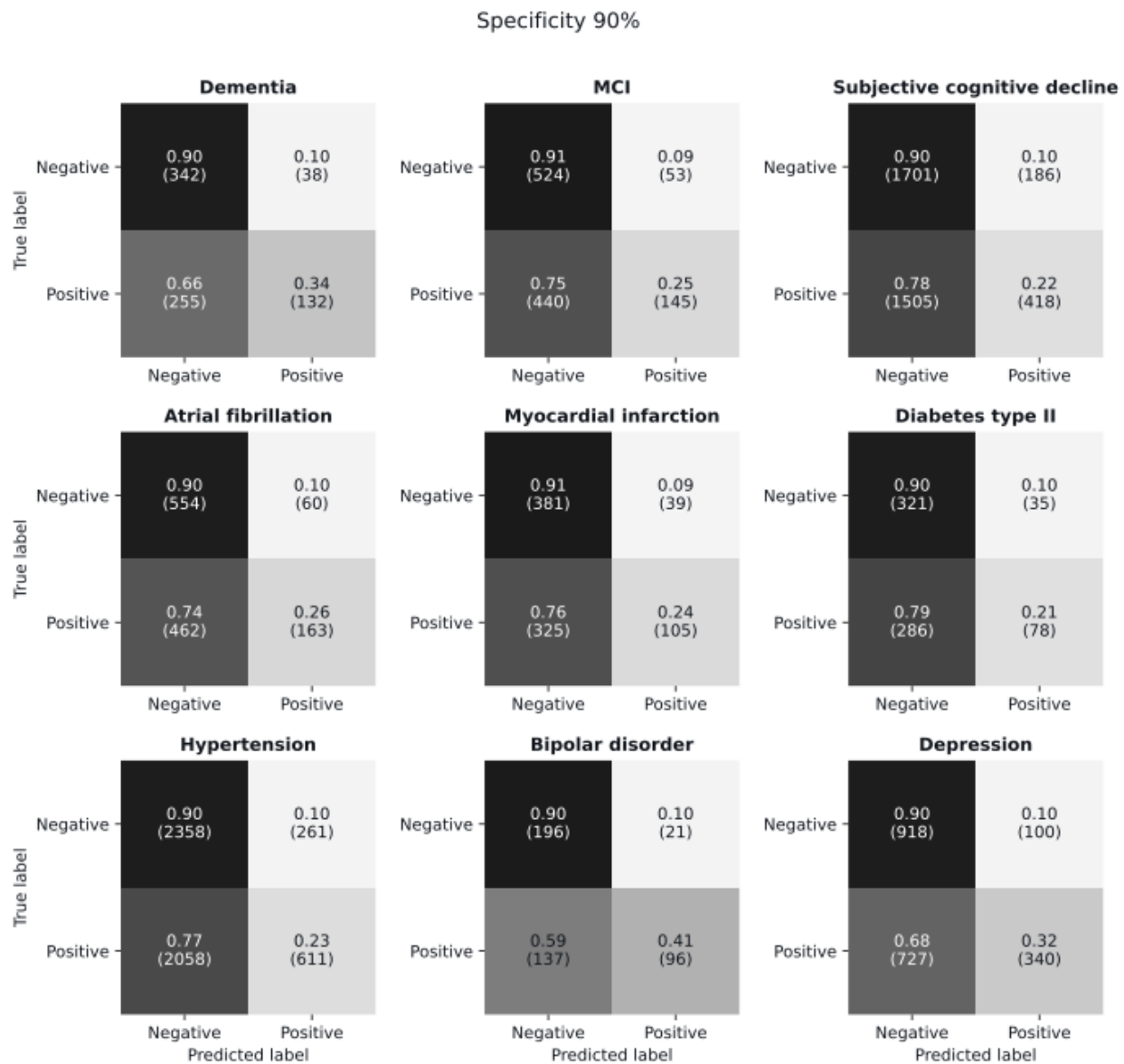


Figure S8. Confusion matrices at specificity 90%

Table S6. Specificity 90% summary table

	Atrial fibrillation	Bipolar disorder	Dementia	Depression	Diabetes type II	Hypertension	MCI	Myocardial infarction	Subjective cognitive decline
Number of samples	1239	450	767	2085	720	5288	1162	850	3810
N positives	625	233	387	1067	364	2669	585	430	1923
N negatives	614	217	380	1018	356	2619	577	420	1887
Prevalence	0.5	0.52	0.5	0.51	0.51	0.5	0.5	0.51	0.5
Threshold	0.73	0.75	0.74	0.77	0.71	0.75	0.73	0.74	0.73
ROC AUC (95% CI)	0.69 (0.66- 0.70)	0.75 (0.73- 0.75)	0.74 (0.73- 0.75)	0.73 (0.71- 0.74)	0.67 (0.63- 0.67)	0.69 (0.69- 0.70)	0.70 (0.69- 0.72)	0.69 (0.68- 0.73)	0.66 (0.64-0.66)
PR AUC (95% CI)	0.66 (0.62- 0.70)	0.77 (0.74- 0.80)	0.73 (0.71- 0.72)	0.69 (0.69- 0.71)	0.64 (0.60- 0.66)	0.65 (0.66- 0.67)	0.68 (0.66- 0.69)	0.67 (0.64- 0.70)	0.63 (0.61-0.64)
Positive Predictive Value	0.73	0.82	0.78	0.77	0.69	0.7	0.73	0.73	0.69
Negative Predictive Value	0.55	0.59	0.57	0.56	0.53	0.53	0.54	0.54	0.53
Predicted Positive Rate	0.18	0.26	0.22	0.21	0.16	0.16	0.17	0.17	0.16

Brain health score – age association

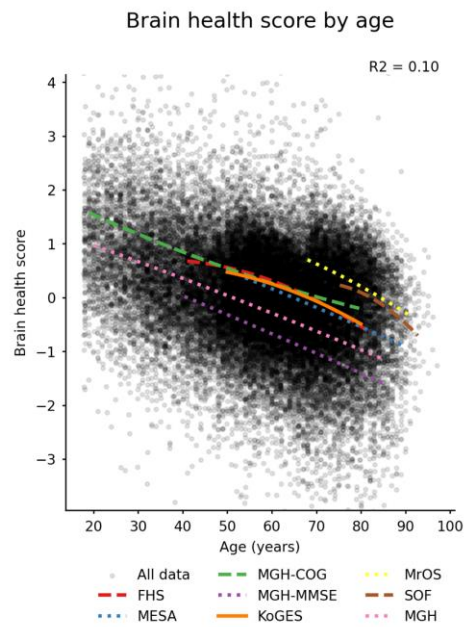


Figure S9: Brain health score – age scatter plot. All sleep recordings across all cohorts are plotted. Quadratic fits per cohort.

Brain health score vs age by cohort

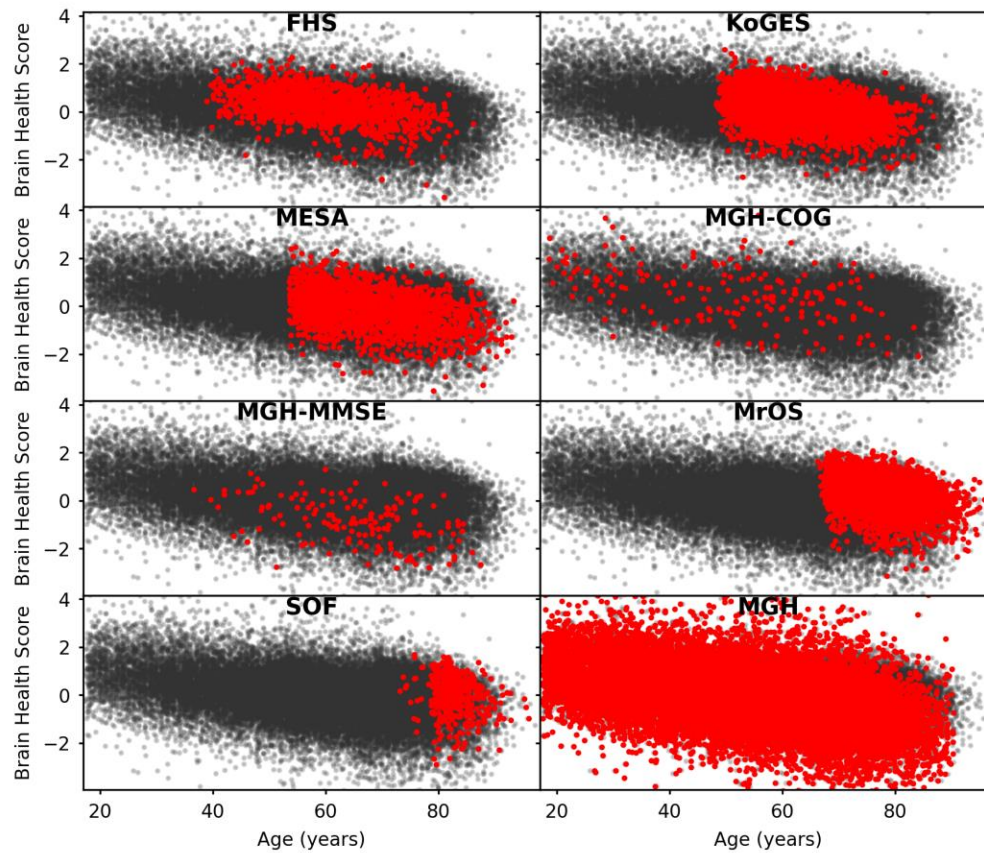


Figure S10. Brain health score – age scatter per cohort. All data across all cohorts are plotted in black, data for individual cohorts plotted in red.

Sensitivity Analysis of the Brain Health Score Optimization Weight (α)

To evaluate how the relative weighting of cognition versus disease targets influences the learned Brain Health Score (BHS), we conducted a post-hoc sensitivity analysis in which we re-trained only the BHS aggregator head while keeping all deep-learning latent representations fixed. This procedure isolates the effect of the α -weighting used in the BHS optimization function without modifying the underlying multi-task representation learned from EEG.

For each of the five cross-validation folds, we loaded the latent features and target variables produced by the full brain health model. We then re-optimized the BHS head across a grid of α values $\in \{0.1, 0.2, \dots, 0.9\}$, where α controls the balance between cognitive and disease objectives:

- Higher α increases emphasis on cognition targets (Huber loss terms).
- Lower α increases emphasis on disease targets (cross-entropy terms).

Each BHS head was trained for 300 epochs with a fixed learning rate (1×10^{-4}), using demographics (age, sex) and latent EEG features as covariates. We evaluated the resulting BHS on held-out fold test sets.

Sensitivity metrics included:

- Pearson correlations between the BHS and cognition across all cohorts
- ROC AUC of the BHS for disease targets
- Mean performance across cognition and disease domains
- Cohort-specific slopes and stability across the α grid

This grid search quantified how shifting the optimization weight affects multi-domain performance and allowed assessment of robustness, stability, and identification of a balanced working point.

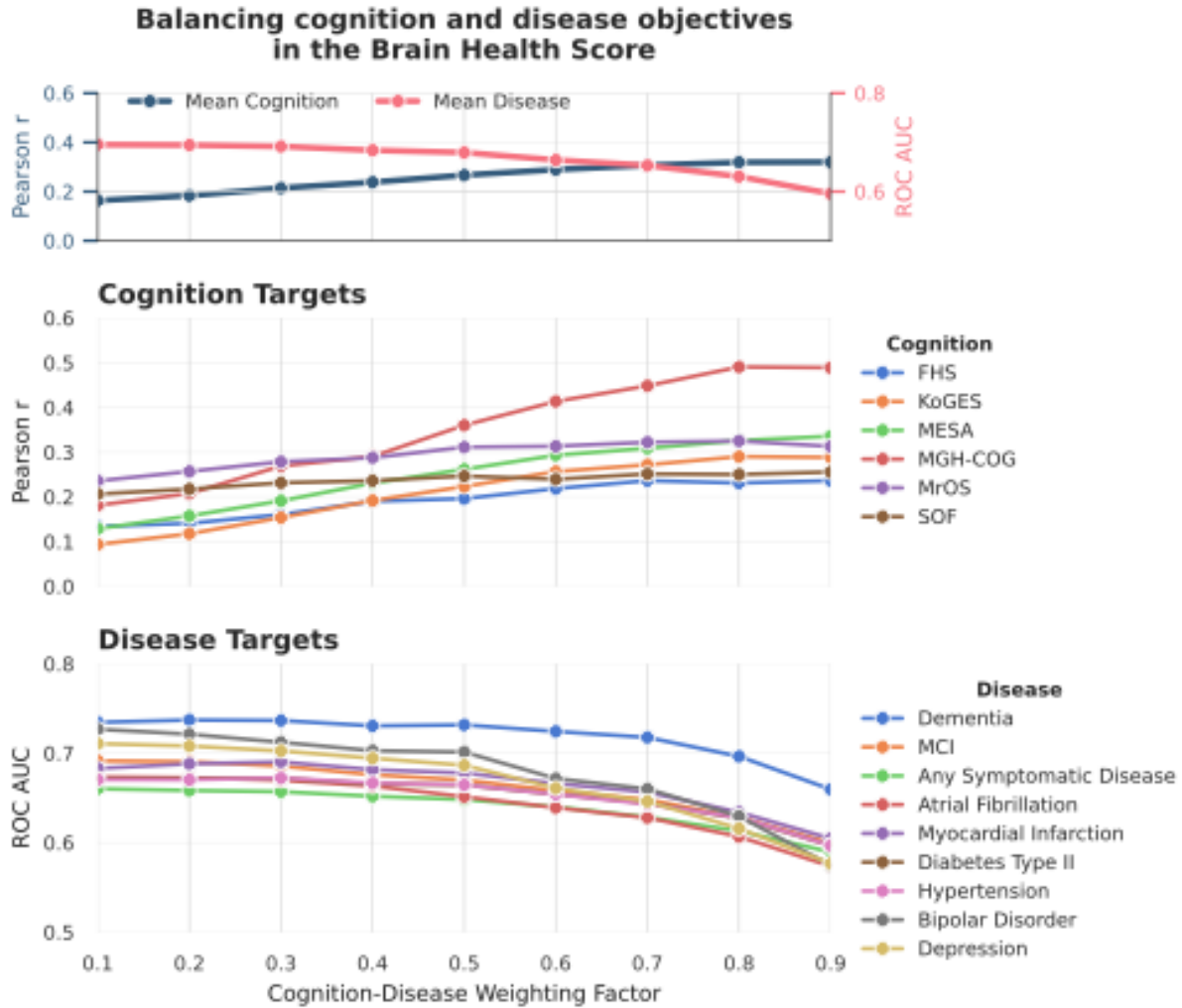


Figure S11. Sensitivity analysis of the Brain Health Score (BHS) optimization weight α . We swept α from 0.1 to 0.9, where higher values increase weighting of cognition and lower values emphasize disease targets in the BHS optimization function. Top: Mean correlation between the BHS and cohort-specific total cognition scores increases monotonically with α , while mean disease ROC AUC decreases smoothly, revealing a stable trade-off between the two domains. Middle: Cognition performance across six cohorts (FHS, KoGES, MESA, MGH-COG, MrOS, SOF) consistently improves as α increases. Bottom: Disease classification AUC for nine diagnostic categories decreases with increasing α , but remains within a high-performing band for $\alpha \leq 0.6$. The chosen working point $\alpha = 0.7$ lies on a natural balance region, achieving near-maximal cognition performance while retaining strong disease discrimination. The smooth, monotonic curves demonstrate that the BHS is robust to a wide range of weighting choices.

Sample spectrograms and error analysis

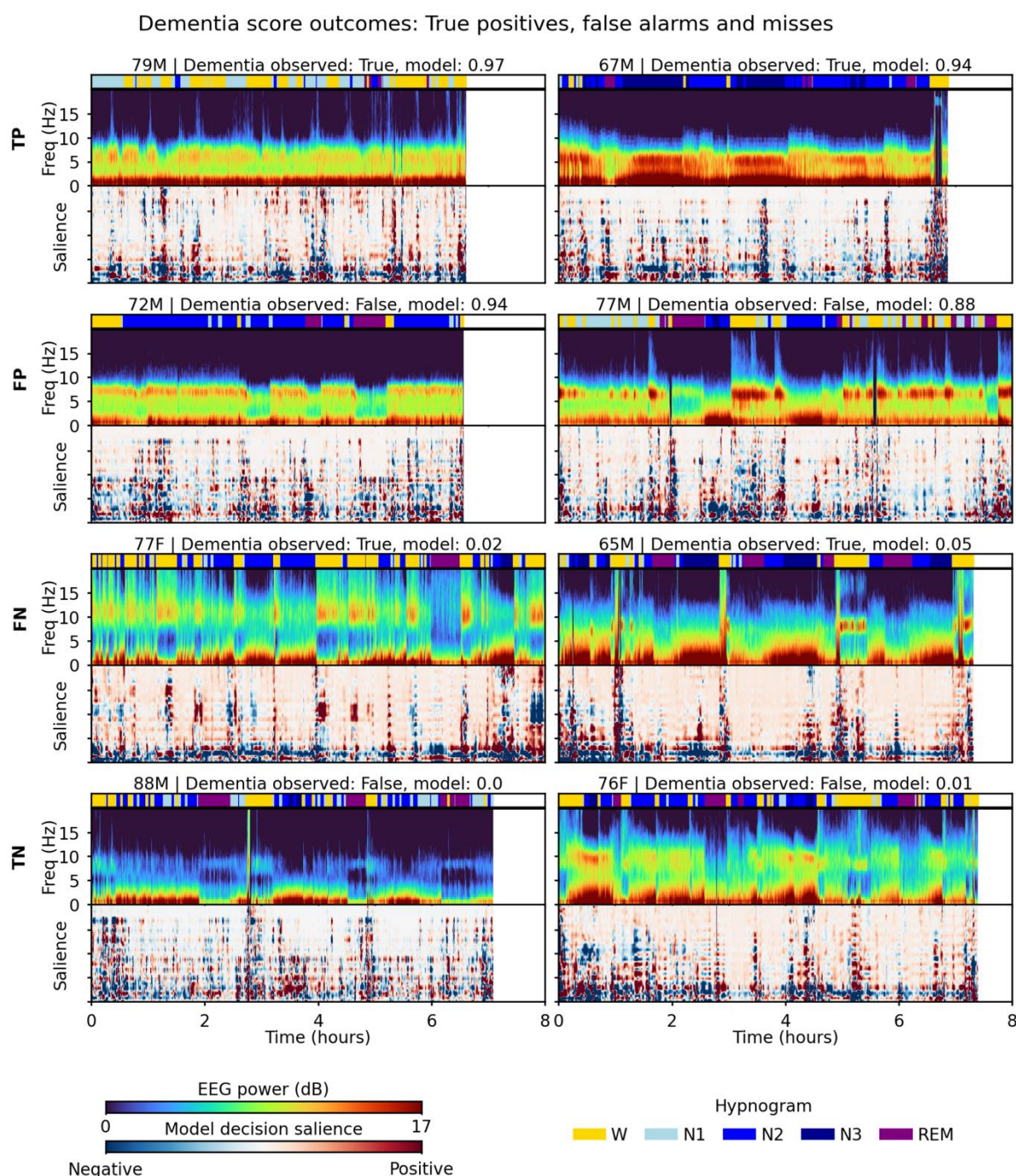


Figure S12: Case-level spectral and saliency patterns underlying dementia predictions: Representative examples illustrating the model’s predictive performance for sampled individuals with low and high dementia score predictions. Each row shows a different outcome category (“true positives,” “false positives,” “false negatives,” and “true negatives”), with the spectrogram

depicting EEG power over time and the bottom panel displaying the Integrated Gradients saliency map (blue-to-red) highlighting features most influential for the model's predictions. The hypnogram tracks sleep stage transitions (W, N1, N2, N3, and REM). Red salience regions indicate positive contributions toward the predicted cognitive score, whereas blue regions suggest negative contributions.

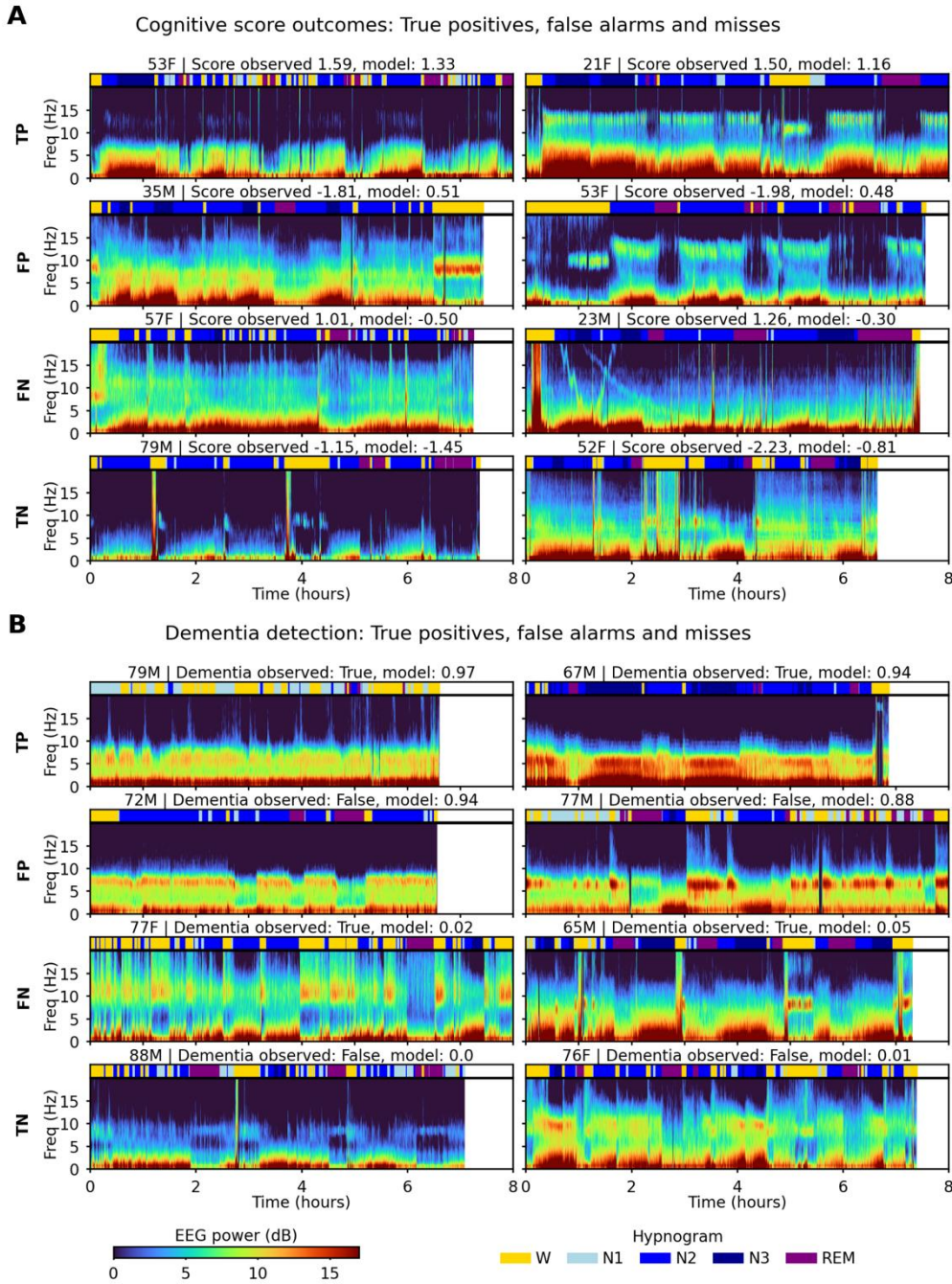


Figure S13: Sample spectrograms and error analysis for cognitive score and dementia prediction.

Auxiliary tasks results

Sleep staging - confusion matrices

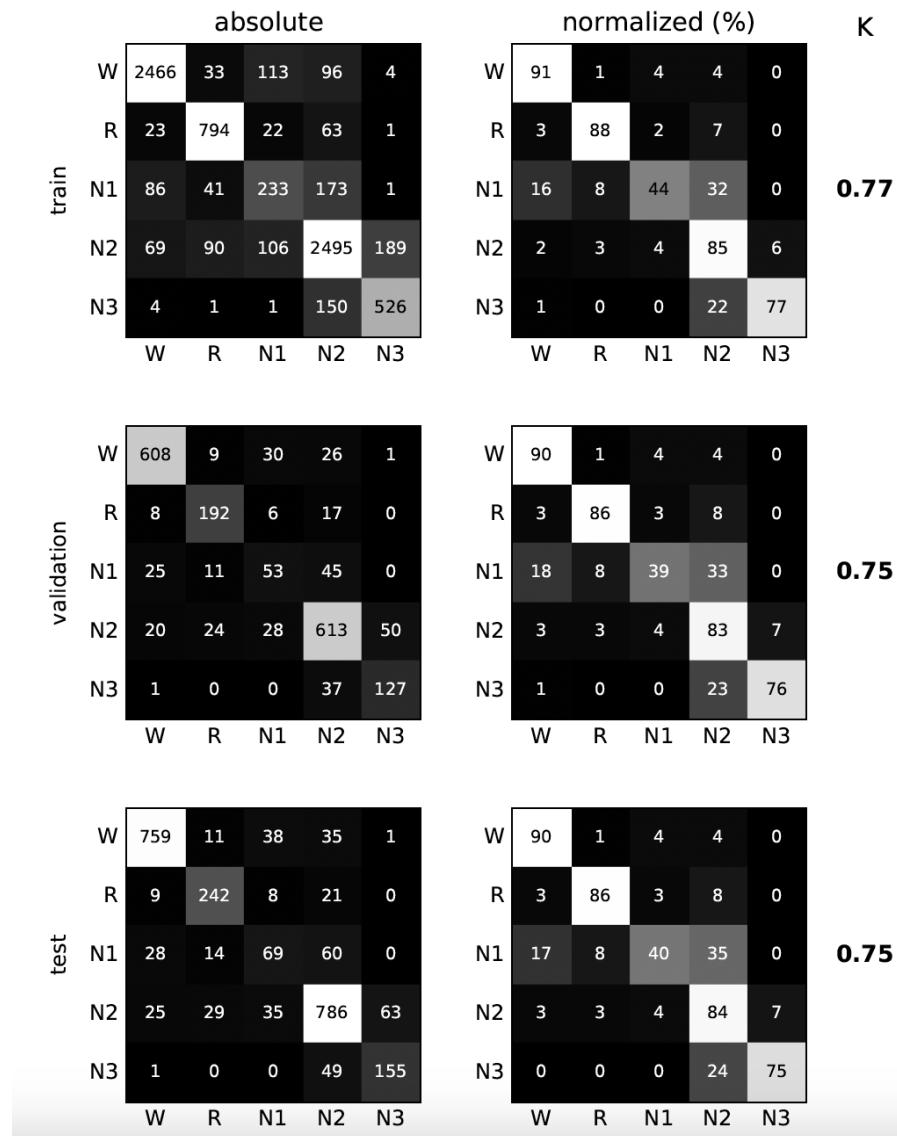


Figure S14. Sleep staging confusion matrices across all cohorts (example fold). Confusion matrices are shown for the training, validation, and test sets. Left panels display absolute numbers of 30-second sleep epochs; right panels display row-normalized percentages (each row corresponds to the true sleep stage, and each column to the predicted stage). Cohen's κ is reported for each split.

Sleep metrics prediction

The “ground truth” values for the sleep-related metrics used in model training were derived from a combination of sources:

1. Expert Annotations: Metrics such as sleep stage proportions (e.g., %N1, %REM) and clinical indices like the Apnea–Hypopnea Index (AHI) were extracted from manually scored PSG reports following standard guidelines.
2. Spectral Calculations: Time-resolved bandpower features (e.g., hourly mean and standard deviation of delta power) were computed directly from the multitaper EEG spectrogram.
3. Event-Based Features: Oscillatory event metrics—including spindle and slow oscillation characteristics—were computed using the open-source LUNA toolkit or our in-house detection algorithms (as detailed in the *Methods* section above).

All target variables were modeled as independent regression tasks, each assigned its own output head in the multi-task neural network. These regression heads shared the same architectural design as those used for cognitive outcomes.

Description of metrics:

1. Frequency Band Metrics

Spectral features were computed within canonical frequency bands: delta (0.5–4 Hz), theta (4–8 Hz), alpha (8–12 Hz), sigma (12–16 Hz), beta (16–30 Hz), and gamma (30–100 Hz). Each band was analyzed using several descriptive statistics:

- `bandpower_max/min/mean/std/kurtosis_c_{stage}`: Maximum, minimum, mean, standard deviation, and kurtosis of band-specific power within a given sleep stage (e.g., n2, n3, rem, w for wake, etc.).
- `bandpower_total / bandpower_4h / bandpower_until_rem`: Total power in a given band across the full night, during the first 4 hours, or prior to first REM onset, respectively.
- `bandpower_{band}-{hour}-mean/std`: Mean or standard deviation of band-specific power within a particular hour of the night (e.g., gamma-8-std denotes gamma power variability in the 8th hour).
- `band ratios` (e.g., `delta/theta_mean` or `slowdelta/fast_mean`): Ratios between power in different bands.

2. Slow Oscillations (SO)

Features prefixed by `so_` characterize slow oscillations (0.3–1 Hz), including:

- `so_rate_c4m1`: Density (events per minute) of SOs across N2 and N3 epochs.
- `so_p2p / so_neg_amp / so_pos_amp`: Peak-to-peak amplitude, negative peak amplitude, and positive peak amplitude of SOs.
- `so_dur / so_neg_dur / so_pos_dur`: Total duration, duration of negative half-wave, and duration of positive half-wave.
- `so_slope_neg1/pos1/neg2/pos2`: Steepness of rising/falling phases of SOs.
- `so_trans / so_trans_freq`: Transition sharpness and oscillatory frequency of SOs.

3. Sleep Spindles (SS)

Spindle features are derived from automatic detection of transient oscillations in the sigma band. Spindles are subclassified as:

- *ss_**: Slow spindles (~11.5 Hz)
- *ms_**: Medium spindles (~13.5 Hz)
- *fs_**: Fast spindles (~15.5 Hz)

For each type, the following metrics were calculated:

- *n / n01 / n02*: Total number of spindles, and counts in first/second halves of the night.
- *dens*: Spindle density per minute.
- *amp, dur, frq, frq1, frq2*: Amplitude, duration, average frequency, and frequency at beginning/end of event.
- *chirp*: Frequency modulation across the spindle.
- *fwhm*: Full-width at half-maximum of the spindle waveform.
- *q*: Spindle quality index based on amplitude-to-noise ratio.
- *symm, symm2*: Temporal symmetry metrics.
- *coupl_angle / coupl_mag / coupl_overlap / coupl_pv*: Phase–amplitude coupling features with slow oscillations, quantifying the timing and strength of spindle–SO interactions.

4. Sleep Architecture & Indices

These are classical PSG features well-known in sleep medicine:

- *AHI* (Apnea-Hypopnea Index): Number of apneas and hypopneas per hour.
- *RDI, OAI, CAI, HYI, RERAI*: Respiratory disturbance indices (e.g., obstructive, central, hypopneas, etc.).
- *WASO*: Wake After Sleep Onset (in minutes).
- *Sleep Efficiency*: Ratio of total sleep time to total recording time.
- *Latency measures (sleep_latency, r_latency)*: Time from lights-off to first N1/REM epoch.
- *SFI* (Sleep Fragmentation Index): Frequency of brief arousals across the night.
- *Perc_{stage}*: Percentage of sleep time in each stage (N1, N2, N3, REM).

Results (Pearson Correlation between “true” and predicted (by deep learning model)):

<i>mean_gradient_c4-m1_w</i>	0.83
<i>kurtosis_c4-m1_w</i>	0.63
<i>slow_bandpower_max_c_w</i>	0.79
<i>delta_bandpower_max_c_w</i>	0.8
<i>slowdelta_bandpower_max_c_w</i>	0.8
<i>theta_bandpower_max_c_w</i>	0.81

<i>alpha_bandpower_max_c_w</i>	0.8
<i>sigma_bandpower_max_c_w</i>	0.78
<i>beta_bandpower_max_c_w</i>	0.81
<i>gamma_bandpower_max_c_w</i>	0.86
<i>slow_bandpower_min_c_w</i>	0.75
<i>delta_bandpower_min_c_w</i>	0.77
<i>slowdelta_bandpower_min_c_w</i>	0.78
<i>theta_bandpower_min_c_w</i>	0.81
<i>alpha_bandpower_min_c_w</i>	0.81
<i>sigma_bandpower_min_c_w</i>	0.79
<i>beta_bandpower_min_c_w</i>	0.82
<i>gamma_bandpower_min_c_w</i>	0.88
<i>slow_bandpower_mean_c_w</i>	0.8
<i>delta_bandpower_mean_c_w</i>	0.8
<i>slowdelta_bandpower_mean_c_w</i>	0.8
<i>theta_bandpower_mean_c_w</i>	0.82
<i>alpha_bandpower_mean_c_w</i>	0.82
<i>sigma_bandpower_mean_c_w</i>	0.79
<i>beta_bandpower_mean_c_w</i>	0.82
<i>gamma_bandpower_mean_c_w</i>	0.88
<i>slow_bandpower_std_c_w</i>	0.79
<i>delta_bandpower_std_c_w</i>	0.8
<i>slowdelta_bandpower_std_c_w</i>	0.8
<i>theta_bandpower_std_c_w</i>	0.8
<i>alpha_bandpower_std_c_w</i>	0.79
<i>sigma_bandpower_std_c_w</i>	0.77
<i>beta_bandpower_std_c_w</i>	0.8
<i>gamma_bandpower_std_c_w</i>	0.85
<i>slow_bandpower_kurtosis_c_w</i>	0.62
<i>delta_bandpower_kurtosis_c_w</i>	0.75
<i>slowdelta_bandpower_kurtosis_c_w</i>	0.73
<i>theta_bandpower_kurtosis_c_w</i>	0.65
<i>alpha_bandpower_kurtosis_c_w</i>	0.53
<i>sigma_bandpower_kurtosis_c_w</i>	0.63
<i>beta_bandpower_kurtosis_c_w</i>	0.68
<i>gamma_bandpower_kurtosis_c_w</i>	0.75
<i>slow/delta_max_c_w</i>	0.76
<i>slow/delta_min_c_w</i>	0.7

<i>slow/delta_mean_c_w</i>	0.75
<i>slow/delta_std_c_w</i>	0.71
<i>delta/theta_max_c_w</i>	0.85
<i>delta/theta_min_c_w</i>	0.84
<i>delta/theta_mean_c_w</i>	0.86
<i>delta/theta_std_c_w</i>	0.84
<i>delta/alpha_max_c_w</i>	0.86
<i>delta/alpha_min_c_w</i>	0.85
<i>delta/alpha_mean_c_w</i>	0.86
<i>delta/alpha_std_c_w</i>	0.85
<i>slowdelta/thetaalpha_max_c_w</i>	0.86
<i>slowdelta/thetaalpha_min_c_w</i>	0.87
<i>slowdelta/thetaalpha_mean_c_w</i>	0.87
<i>slowdelta/thetaalpha_std_c_w</i>	0.85
<i>slowdelta/fast_max_c_w</i>	0.83
<i>slowdelta/fast_min_c_w</i>	0.78
<i>slowdelta/fast_mean_c_w</i>	0.84
<i>slowdelta/fast_std_c_w</i>	0.83
<i>theta/alpha_max_c_w</i>	0.84
<i>theta/alpha_min_c_w</i>	0.84
<i>theta/alpha_mean_c_w</i>	0.85
<i>theta/alpha_std_c_w</i>	0.82
<i>mean_gradient_c4-ml_r</i>	0.8
<i>kurtosis_c4-ml_r</i>	0.53
<i>slow_bandpower_max_c_r</i>	0.67
<i>delta_bandpower_max_c_r</i>	0.64
<i>slowdelta_bandpower_max_c_r</i>	0.65
<i>theta_bandpower_max_c_r</i>	0.73
<i>alpha_bandpower_max_c_r</i>	0.75
<i>sigma_bandpower_max_c_r</i>	0.69
<i>beta_bandpower_max_c_r</i>	0.65
<i>gamma_bandpower_max_c_r</i>	0.67
<i>slow_bandpower_min_c_r</i>	0.66
<i>delta_bandpower_min_c_r</i>	0.68
<i>slowdelta_bandpower_min_c_r</i>	0.68
<i>theta_bandpower_min_c_r</i>	0.74
<i>alpha_bandpower_min_c_r</i>	0.73

<i>sigma_bandpower_min_c_r</i>	0.75
<i>beta_bandpower_min_c_r</i>	0.76
<i>gamma_bandpower_min_c_r</i>	0.77
<i>slow_bandpower_mean_c_r</i>	0.69
<i>delta_bandpower_mean_c_r</i>	0.67
<i>slowdelta_bandpower_mean_c_r</i>	0.68
<i>theta_bandpower_mean_c_r</i>	0.74
<i>alpha_bandpower_mean_c_r</i>	0.74
<i>sigma_bandpower_mean_c_r</i>	0.75
<i>beta_bandpower_mean_c_r</i>	0.71
<i>gamma_bandpower_mean_c_r</i>	0.75
<i>slow_bandpower_std_c_r</i>	0.67
<i>delta_bandpower_std_c_r</i>	0.62
<i>slowdelta_bandpower_std_c_r</i>	0.62
<i>theta_bandpower_std_c_r</i>	0.7
<i>alpha_bandpower_std_c_r</i>	0.73
<i>sigma_bandpower_std_c_r</i>	0.64
<i>beta_bandpower_std_c_r</i>	0.6
<i>gamma_bandpower_std_c_r</i>	0.63
<i>slow_bandpower_kurtosis_c_r</i>	0.52
<i>delta_bandpower_kurtosis_c_r</i>	0.73
<i>slowdelta_bandpower_kurtosis_c_r</i>	0.7
<i>theta_bandpower_kurtosis_c_r</i>	0.5
<i>alpha_bandpower_kurtosis_c_r</i>	0.56
<i>sigma_bandpower_kurtosis_c_r</i>	0.58
<i>beta_bandpower_kurtosis_c_r</i>	0.63
<i>gamma_bandpower_kurtosis_c_r</i>	0.51
<i>slow/delta_max_c_r</i>	0.82
<i>slow/delta_min_c_r</i>	0.7
<i>slow/delta_mean_c_r</i>	0.8
<i>slow/delta_std_c_r</i>	0.78
<i>delta/theta_max_c_r</i>	0.74
<i>delta/theta_min_c_r</i>	0.73
<i>delta/theta_mean_c_r</i>	0.76
<i>delta/theta_std_c_r</i>	0.73

<i>delta/alpha_max_c_r</i>	0.76
<i>delta/alpha_min_c_r</i>	0.77
<i>delta/alpha_mean_c_r</i>	0.77
<i>delta/alpha_std_c_r</i>	0.73
<i>slowdelta/thetaalpha_max_c_r</i>	0.77
<i>slowdelta/thetaalpha_min_c_r</i>	0.77
<i>slowdelta/thetaalpha_mean_c_r</i>	0.79
<i>slowdelta/thetaalpha_std_c_r</i>	0.77
<i>slowdelta/fast_max_c_r</i>	0.78
<i>slowdelta/fast_min_c_r</i>	0.81
<i>slowdelta/fast_mean_c_r</i>	0.8
<i>slowdelta/fast_std_c_r</i>	0.77
<i>theta/alpha_max_c_r</i>	0.73
<i>theta/alpha_min_c_r</i>	0.77
<i>theta/alpha_mean_c_r</i>	0.76
<i>theta/alpha_std_c_r</i>	0.73
<i>mean_gradient_c4-m1_n1</i>	0.74
<i>kurtosis_c4-m1_n1</i>	0.57
<i>slow_bandpower_max_c_n1</i>	0.66
<i>delta_bandpower_max_c_n1</i>	0.67
<i>slowdelta_bandpower_max_c_n1</i>	0.67
<i>theta_bandpower_max_c_n1</i>	0.74
<i>alpha_bandpower_max_c_n1</i>	0.73
<i>sigma_bandpower_max_c_n1</i>	0.66
<i>beta_bandpower_max_c_n1</i>	0.63
<i>gamma_bandpower_max_c_n1</i>	0.65
<i>slow_bandpower_min_c_n1</i>	0.63
<i>delta_bandpower_min_c_n1</i>	0.68
<i>slowdelta_bandpower_min_c_n1</i>	0.68
<i>theta_bandpower_min_c_n1</i>	0.72
<i>alpha_bandpower_min_c_n1</i>	0.72
<i>sigma_bandpower_min_c_n1</i>	0.7
<i>beta_bandpower_min_c_n1</i>	0.65
<i>gamma_bandpower_min_c_n1</i>	0.72
<i>slow_bandpower_mean_c_n1</i>	0.69

<i>delta_bandpower_mean_c_n1</i>	0.68
<i>slowdelta_bandpower_mean_c_n1</i>	0.69
<i>theta_bandpower_mean_c_n1</i>	0.75
<i>alpha_bandpower_mean_c_n1</i>	0.76
<i>sigma_bandpower_mean_c_n1</i>	0.68
<i>beta_bandpower_mean_c_n1</i>	0.65
<i>gamma_bandpower_mean_c_n1</i>	0.67
<i>slow_bandpower_std_c_n1</i>	0.66
<i>delta_bandpower_std_c_n1</i>	0.65
<i>slowdelta_bandpower_std_c_n1</i>	0.67
<i>theta_bandpower_std_c_n1</i>	0.74
<i>alpha_bandpower_std_c_n1</i>	0.73
<i>sigma_bandpower_std_c_n1</i>	0.62
<i>beta_bandpower_std_c_n1</i>	0.62
<i>gamma_bandpower_std_c_n1</i>	0.65
<i>slow_bandpower_kurtosis_c_n1</i>	0.43
<i>delta_bandpower_kurtosis_c_n1</i>	0.65
<i>slowdelta_bandpower_kurtosis_c_n1</i>	0.61
<i>theta_bandpower_kurtosis_c_n1</i>	0.56
<i>alpha_bandpower_kurtosis_c_n1</i>	0.52
<i>sigma_bandpower_kurtosis_c_n1</i>	0.5
<i>beta_bandpower_kurtosis_c_n1</i>	0.6
<i>gamma_bandpower_kurtosis_c_n1</i>	0.62
<i>slow/delta_max_c_n1</i>	0.75
<i>slow/delta_min_c_n1</i>	0.65
<i>slow/delta_mean_c_n1</i>	0.75
<i>slow/delta_std_c_n1</i>	0.69
<i>delta/theta_max_c_n1</i>	0.73
<i>delta/theta_min_c_n1</i>	0.74
<i>delta/theta_mean_c_n1</i>	0.76
<i>delta/theta_std_c_n1</i>	0.7
<i>delta/alpha_max_c_n1</i>	0.73
<i>delta/alpha_min_c_n1</i>	0.72
<i>delta/alpha_mean_c_n1</i>	0.75
<i>delta/alpha_std_c_n1</i>	0.73

<i>slowdelta/thetaalpha_max_c_n1</i>	0.74
<i>slowdelta/thetaalpha_min_c_n1</i>	0.74
<i>slowdelta/thetaalpha_mean_c_n1</i>	0.76
<i>slowdelta/thetaalpha_std_c_n1</i>	0.74
<i>slowdelta/fast_max_c_n1</i>	0.71
<i>slowdelta/fast_min_c_n1</i>	0.67
<i>slowdelta/fast_mean_c_n1</i>	0.71
<i>slowdelta/fast_std_c_n1</i>	0.71
<i>theta/alpha_max_c_n1</i>	0.78
<i>theta/alpha_min_c_n1</i>	0.8
<i>theta/alpha_mean_c_n1</i>	0.8
<i>theta/alpha_std_c_n1</i>	0.77
<i>mean_gradient_c4-m1_n2</i>	0.82
<i>kurtosis_c4-m1_n2</i>	0.69
<i>slow_bandpower_max_c_n2</i>	0.82
<i>delta_bandpower_max_c_n2</i>	0.8
<i>slowdelta_bandpower_max_c_n2</i>	0.81
<i>theta_bandpower_max_c_n2</i>	0.81
<i>alpha_bandpower_max_c_n2</i>	0.83
<i>sigma_bandpower_max_c_n2</i>	0.83
<i>beta_bandpower_max_c_n2</i>	0.77
<i>gamma_bandpower_max_c_n2</i>	0.82
<i>slow_bandpower_min_c_n2</i>	0.74
<i>delta_bandpower_min_c_n2</i>	0.72
<i>slowdelta_bandpower_min_c_n2</i>	0.71
<i>theta_bandpower_min_c_n2</i>	0.78
<i>alpha_bandpower_min_c_n2</i>	0.81
<i>sigma_bandpower_min_c_n2</i>	0.82
<i>beta_bandpower_min_c_n2</i>	0.83
<i>gamma_bandpower_min_c_n2</i>	0.86
<i>slow_bandpower_mean_c_n2</i>	0.81
<i>delta_bandpower_mean_c_n2</i>	0.79
<i>slowdelta_bandpower_mean_c_n2</i>	0.79
<i>theta_bandpower_mean_c_n2</i>	0.8
<i>alpha_bandpower_mean_c_n2</i>	0.83

<i>sigma_bandpower_mean_c_n2</i>	0.82
<i>beta_bandpower_mean_c_n2</i>	0.78
<i>gamma_bandpower_mean_c_n2</i>	0.87
<i>slow_bandpower_std_c_n2</i>	0.81
<i>delta_bandpower_std_c_n2</i>	0.81
<i>slowdelta_bandpower_std_c_n2</i>	0.81
<i>theta_bandpower_std_c_n2</i>	0.81
<i>alpha_bandpower_std_c_n2</i>	0.83
<i>sigma_bandpower_std_c_n2</i>	0.83
<i>beta_bandpower_std_c_n2</i>	0.75
<i>gamma_bandpower_std_c_n2</i>	0.82
<i>slow_bandpower_kurtosis_c_n2</i>	0.65
<i>delta_bandpower_kurtosis_c_n2</i>	0.77
<i>slowdelta_bandpower_kurtosis_c_n2</i>	0.73
<i>theta_bandpower_kurtosis_c_n2</i>	0.73
<i>alpha_bandpower_kurtosis_c_n2</i>	0.68
<i>sigma_bandpower_kurtosis_c_n2</i>	0.72
<i>beta_bandpower_kurtosis_c_n2</i>	0.8
<i>gamma_bandpower_kurtosis_c_n2</i>	0.66
<i>slow/delta_max_c_n2</i>	0.9
<i>slow/delta_min_c_n2</i>	0.85
<i>slow/delta_mean_c_n2</i>	0.89
<i>slow/delta_std_c_n2</i>	0.85
<i>delta/theta_max_c_n2</i>	0.86
<i>delta/theta_min_c_n2</i>	0.82
<i>delta/theta_mean_c_n2</i>	0.85
<i>delta/theta_std_c_n2</i>	0.85
<i>delta/alpha_max_c_n2</i>	0.87
<i>delta/alpha_min_c_n2</i>	0.86
<i>delta/alpha_mean_c_n2</i>	0.88
<i>delta/alpha_std_c_n2</i>	0.87
<i>slowdelta/thetaalpha_max_c_n2</i>	0.87
<i>slowdelta/thetaalpha_min_c_n2</i>	0.84
<i>slowdelta/thetaalpha_mean_c_n2</i>	0.87
<i>slowdelta/thetaalpha_std_c_n2</i>	0.87

<i>slowdelta/fast_max_c_n2</i>	0.86
<i>slowdelta/fast_min_c_n2</i>	0.83
<i>slowdelta/fast_mean_c_n2</i>	0.86
<i>slowdelta/fast_std_c_n2</i>	0.86
<i>theta/alpha_max_c_n2</i>	0.88
<i>theta/alpha_min_c_n2</i>	0.9
<i>theta/alpha_mean_c_n2</i>	0.9
<i>theta/alpha_std_c_n2</i>	0.87
<i>mean_gradient_c4-m1_n3</i>	0.78
<i>kurtosis_c4-m1_n3</i>	0.45
<i>slow_bandpower_max_c_n3</i>	0.74
<i>delta_bandpower_max_c_n3</i>	0.76
<i>slowdelta_bandpower_max_c_n3</i>	0.75
<i>theta_bandpower_max_c_n3</i>	0.74
<i>alpha_bandpower_max_c_n3</i>	0.76
<i>sigma_bandpower_max_c_n3</i>	0.74
<i>beta_bandpower_max_c_n3</i>	0.62
<i>gamma_bandpower_max_c_n3</i>	0.64
<i>slow_bandpower_min_c_n3</i>	0.7
<i>delta_bandpower_min_c_n3</i>	0.7
<i>slowdelta_bandpower_min_c_n3</i>	0.69
<i>theta_bandpower_min_c_n3</i>	0.72
<i>alpha_bandpower_min_c_n3</i>	0.76
<i>sigma_bandpower_min_c_n3</i>	0.74
<i>beta_bandpower_min_c_n3</i>	0.73
<i>gamma_bandpower_min_c_n3</i>	0.71
<i>slow_bandpower_mean_c_n3</i>	0.76
<i>delta_bandpower_mean_c_n3</i>	0.74
<i>slowdelta_bandpower_mean_c_n3</i>	0.73
<i>theta_bandpower_mean_c_n3</i>	0.75
<i>alpha_bandpower_mean_c_n3</i>	0.76
<i>sigma_bandpower_mean_c_n3</i>	0.74
<i>beta_bandpower_mean_c_n3</i>	0.73
<i>gamma_bandpower_mean_c_n3</i>	0.75
<i>slow_bandpower_std_c_n3</i>	0.73

<i>delta_bandpower_std_c_n3</i>	0.75
<i>slowdelta_bandpower_std_c_n3</i>	0.75
<i>theta_bandpower_std_c_n3</i>	0.73
<i>alpha_bandpower_std_c_n3</i>	0.76
<i>sigma_bandpower_std_c_n3</i>	0.75
<i>beta_bandpower_std_c_n3</i>	0.53
<i>gamma_bandpower_std_c_n3</i>	0.55
<i>slow_bandpower_kurtosis_c_n3</i>	0.27
<i>delta_bandpower_kurtosis_c_n3</i>	0.59
<i>slowdelta_bandpower_kurtosis_c_n3</i>	0.5
<i>theta_bandpower_kurtosis_c_n3</i>	0.58
<i>alpha_bandpower_kurtosis_c_n3</i>	0.46
<i>sigma_bandpower_kurtosis_c_n3</i>	0.62
<i>beta_bandpower_kurtosis_c_n3</i>	0.7
<i>gamma_bandpower_kurtosis_c_n3</i>	0.57
<i>slow/delta_max_c_n3</i>	0.82
<i>slow/delta_min_c_n3</i>	0.74
<i>slow/delta_mean_c_n3</i>	0.82
<i>slow/delta_std_c_n3</i>	0.67
<i>delta/theta_max_c_n3</i>	0.78
<i>delta/theta_min_c_n3</i>	0.74
<i>delta/theta_mean_c_n3</i>	0.8
<i>delta/theta_std_c_n3</i>	0.77
<i>delta/alpha_max_c_n3</i>	0.83
<i>delta/alpha_min_c_n3</i>	0.81
<i>delta/alpha_mean_c_n3</i>	0.84
<i>delta/alpha_std_c_n3</i>	0.81
<i>slowdelta/thetaalpha_max_c_n3</i>	0.8
<i>slowdelta/thetaalpha_min_c_n3</i>	0.78
<i>slowdelta/thetaalpha_mean_c_n3</i>	0.82
<i>slowdelta/thetaalpha_std_c_n3</i>	0.79
<i>slowdelta/fast_max_c_n3</i>	0.83
<i>slowdelta/fast_min_c_n3</i>	0.81
<i>slowdelta/fast_mean_c_n3</i>	0.83
<i>slowdelta/fast_std_c_n3</i>	0.81

<i>theta/alpha_max_c_n3</i>	0.83
<i>theta/alpha_min_c_n3</i>	0.83
<i>theta/alpha_mean_c_n3</i>	0.84
<i>theta/alpha_std_c_n3</i>	0.81
<i>delta_linear_b1</i>	0.59
<i>delta_linear_b0</i>	0.76
<i>slowdelta_bandpower_total</i>	0.84
<i>slowdelta_bandpower_4h</i>	0.83
<i>slowdelta_bandpower_until_rem</i>	0.68
<i>first_rem_duration</i>	0.4
<i>first_rem_slowdelta_bandpower_mean_c</i>	0.42
<i>first_rem_theta_bandpower_mean_c</i>	0.5
<i>first_rem_alpha_bandpower_mean_c</i>	0.5
<i>first_rem_fastband_bandpower_mean_c</i>	0.42
<i>remaining_rem_duration</i>	0.49
<i>remaining_rem_slowdelta_bandpower_mean_c</i>	0.49
<i>remaining_rem_theta_bandpower_mean_c</i>	0.55
<i>remaining_rem_alpha_bandpower_mean_c</i>	0.58
<i>remaining_rem_fastband_bandpower_mean_c</i>	0.51
<i>hours_sleep</i>	0.8
<i>hours_psg</i>	0.92
<i>sleep_efficiency</i>	0.88
<i>perc_r</i>	0.58
<i>perc_n1</i>	0.81
<i>perc_n2</i>	0.65
<i>perc_n3</i>	0.68
<i>waso</i>	0.84
<i>sleep_latency</i>	0.73
<i>r_latency</i>	0.55
<i>sfi</i>	0.51
<i>ahi</i>	0.58
<i>rdi</i>	0.53
<i>oai</i>	0.34
<i>cai</i>	0.4
<i>hyi</i>	0.62

<i>rera</i>	0.53
<i>arousal_index</i>	0.53
<i>ss_amp_c4m1</i>	0.71
<i>ss_chirp_c4m1</i>	0.52
<i>ss_dens_c4m1</i>	0.74
<i>ss_isa_m_c4m1</i>	0.62
<i>ss_isa_s_c4m1</i>	0.55
<i>ss_isa_t_c4m1</i>	0.63
<i>ss_n_c4m1</i>	0.74
<i>ss_n01_c4m1</i>	0.7
<i>ss_n02_c4m1</i>	0.71
<i>ss_nosc_c4m1</i>	0.58
<i>ss_q_c4m1</i>	0.62
<i>ss_symm_c4m1</i>	0.25
<i>ss_symm2_c4m1</i>	0.27
<i>ss_fwhm_c4m1</i>	0.59
<i>ss_mins_c4m1</i>	0.69
<i>ss_frq_c4m1</i>	0.8
<i>ss_frq1_c4m1</i>	0.8
<i>ss_frq2_c4m1</i>	0.78
<i>ss_dur_c4m1</i>	0.43
<i>ss_fft_c4m1</i>	0.78
<i>ss_dispersion_c4m1</i>	0.29
<i>ss_dispersion_p_c4m1</i>	0.21
<i>ss_coupl_angle_c4m1</i>	0.38
<i>ss_coupl_mag_c4m1</i>	0.19
<i>ss_coupl_overlap_c4m1</i>	0.66
<i>ss_coupl_pv_c4m1</i>	0.41
<i>ms_amp_c4m1</i>	0.72
<i>ms_chirp_c4m1</i>	0.49
<i>ms_dens_c4m1</i>	0.78
<i>ms_isa_m_c4m1</i>	0.74
<i>ms_isa_s_c4m1</i>	0.55
<i>ms_isa_t_c4m1</i>	0.73
<i>ms_n_c4m1</i>	0.76

<i>ms_n01_c4m1</i>	0.72
<i>ms_n02_c4m1</i>	0.74
<i>ms_nosc_c4m1</i>	0.64
<i>ms_q_c4m1</i>	0.6
<i>ms_symm_c4m1</i>	0.27
<i>ms_symm2_c4m1</i>	0.36
<i>ms_fwhm_c4m1</i>	0.73
<i>ms_mins_c4m1</i>	0.69
<i>ms_frq_c4m1</i>	0.74
<i>ms_frq1_c4m1</i>	0.74
<i>ms_frq2_c4m1</i>	0.72
<i>ms_dur_c4m1</i>	0.62
<i>ms_fft_c4m1</i>	0.7
<i>ms_dispersion_c4m1</i>	0.31
<i>ms_dispersion_p_c4m1</i>	0.19
<i>ms_coupl_angle_c4m1</i>	0.58
<i>ms_coupl_mag_c4m1</i>	0.33
<i>ms_coupl_overlap_c4m1</i>	0.66
<i>ms_coupl_pv_c4m1</i>	0.44
<i>fs_amp_c4m1</i>	0.68
<i>fs_chirp_c4m1</i>	0.35
<i>fs_dens_c4m1</i>	0.78
<i>fs_isa_m_c4m1</i>	0.73
<i>fs_isa_s_c4m1</i>	0.52
<i>fs_isa_t_c4m1</i>	0.72
<i>fs_n_c4m1</i>	0.81
<i>fs_n01_c4m1</i>	0.75
<i>fs_n02_c4m1</i>	0.8
<i>fs_nosc_c4m1</i>	0.47
<i>fs_q_c4m1</i>	0.61
<i>fs_symm_c4m1</i>	0.31
<i>fs_symm2_c4m1</i>	0.46
<i>fs_fwhm_c4m1</i>	0.7
<i>fs_mins_c4m1</i>	0.68
<i>fs_frq_c4m1</i>	0.72

<i>fs_frq1_c4m1</i>	0.73
<i>fs_frq2_c4m1</i>	0.74
<i>fs_dur_c4m1</i>	0.48
<i>fs_fft_c4m1</i>	0.7
<i>fs_dispersion_c4m1</i>	0.27
<i>fs_dispersion_p_c4m1</i>	0.28
<i>fs_coupl_angle_c4m1</i>	0.53
<i>fs_coupl_mag_c4m1</i>	0.24
<i>fs_coupl_overlap_c4m1</i>	0.68
<i>fs_coupl_pv_c4m1</i>	0.51
<i>so_c4m1</i>	0.85
<i>so_dur_c4m1</i>	0.8
<i>so_neg_amp_c4m1</i>	0.74
<i>so_neg_dur_c4m1</i>	0.78
<i>so_p2p_c4m1</i>	0.77
<i>so_pos_amp_c4m1</i>	0.77
<i>so_pos_dur_c4m1</i>	0.8
<i>so_rate_c4m1</i>	0.84
<i>so_slope_neg1_c4m1</i>	0.73
<i>so_slope_neg2_c4m1</i>	0.71
<i>so_slope_pos1_c4m1</i>	0.68
<i>so_slope_pos2_c4m1</i>	0.76
<i>so_trans_c4m1</i>	0.74
<i>so_trans_freq_c4m1</i>	0.74
<i>total_power_full_sleep_abs_so</i>	0.84
<i>total_power_full_sleep_rel_so</i>	0.81
<i>total_power_full_sleep_rel_to_total_so</i>	0.76
<i>total_power_rem_abs_so</i>	0.66
<i>total_power_rem_rel_so</i>	0.74
<i>total_power_rem_rel_to_total_so</i>	0.7
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<i>total_power_n1_rel_so</i>	0.76

<i>total_power_n1_rel_to_total_so</i>	0.81
<i>total_power_n2_abs_so</i>	0.82
<i>total_power_n2_rel_so</i>	0.78
<i>total_power_n2_rel_to_total_so</i>	0.67
<i>total_power_n3_abs_so</i>	0.7
<i>total_power_n3_rel_so</i>	0.76
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<i>total_power_first_stable_rem_rel_to_total_so</i>	0.71
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<i>total_power_rem_rel_delta</i>	0.67
<i>total_power_rem_rel_to_total_delta</i>	0.78
<i>total_power_nrem_abs_delta</i>	0.81
<i>total_power_nrem_rel_delta</i>	0.8
<i>total_power_nrem_rel_to_total_delta</i>	0.89
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<i>total_power_n1_rel_delta</i>	0.65
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<i>total_power_n2_abs_delta</i>	0.8
<i>total_power_n2_rel_delta</i>	0.77
<i>total_power_n2_rel_to_total_delta</i>	0.83
<i>total_power_n3_abs_delta</i>	0.71
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<i>total_power_w_abs_delta</i>	0.92

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<i>total_power_nrem_abs_theta</i>	0.78
<i>total_power_nrem_rel_theta</i>	0.85
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<i>total_power_n1_rel_to_total_theta</i>	0.88
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<i>total_power_n2_rel_theta</i>	0.84
<i>total_power_n2_rel_to_total_theta</i>	0.87
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<i>total_power_n3_rel_theta</i>	0.78
<i>total_power_n3_rel_to_total_theta</i>	0.86
<i>total_power_w_abs_theta</i>	0.92
<i>total_power_w_rel_theta</i>	0.77
<i>total_power_w_rel_to_total_theta</i>	0.82
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<i>total_power_first_stable_nrem_rel_theta</i>	0.74
<i>total_power_first_stable_nrem_rel_to_total_theta</i>	0.89
<i>total_power_first_stable_rem_abs_theta</i>	0.51
<i>total_power_first_stable_rem_rel_theta</i>	0.68
<i>total_power_first_stable_rem_rel_to_total_theta</i>	0.83

<i>total_power_full_sleep_abs_alpha</i>	0.77
<i>total_power_full_sleep_rel_alpha</i>	0.86
<i>total_power_full_sleep_rel_to_total_alpha</i>	0.9
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<i>total_power_rem_rel_alpha</i>	0.74
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<i>total_power_first_stable_rem_rel_to_total_alpha</i>	0.82
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<i>total_power_rem_rel_sigma</i>	0.72
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<i>total_power_nrem_rel_sigma</i>	0.84

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<i>total_power_n3_abs_sigma</i>	0.63
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<i>total_power_w_abs_sigma</i>	0.92
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<i>total_power_n1_abs_beta</i>	0.73
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<i>so-2-std</i>	0.65
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<i>so-3-std</i>	0.67
<i>so-4-mean</i>	0.73
<i>so-4-std</i>	0.72
<i>so-5-mean</i>	0.7
<i>so-5-std</i>	0.7
<i>so-6-mean</i>	0.66
<i>so-6-std</i>	0.66
<i>so-7-mean</i>	0.78
<i>so-7-std</i>	0.78
<i>so-8-mean</i>	0.9
<i>so-8-std</i>	0.9
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<i>delta-0-std</i>	0.67
<i>delta-1-mean</i>	0.65
<i>delta-1-std</i>	0.63
<i>delta-2-mean</i>	0.63
<i>delta-2-std</i>	0.64
<i>delta-3-mean</i>	0.66
<i>delta-3-std</i>	0.66
<i>delta-4-mean</i>	0.7
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<i>delta-5-mean</i>	0.68
<i>delta-5-std</i>	0.68
<i>delta-6-mean</i>	0.64

<i>delta-6-std</i>	0.64
<i>delta-7-mean</i>	0.79
<i>delta-7-std</i>	0.78
<i>delta-8-mean</i>	0.9
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<i>theta-2-std</i>	0.65
<i>theta-3-mean</i>	0.7
<i>theta-3-std</i>	0.66
<i>theta-4-mean</i>	0.73
<i>theta-4-std</i>	0.69
<i>theta-5-mean</i>	0.72
<i>theta-5-std</i>	0.67
<i>theta-6-mean</i>	0.67
<i>theta-6-std</i>	0.62
<i>theta-7-mean</i>	0.79
<i>theta-7-std</i>	0.78
<i>theta-8-mean</i>	0.9
<i>theta-8-std</i>	0.9
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<i>alpha-1-mean</i>	0.61
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<i>alpha-2-mean</i>	0.65
<i>alpha-2-std</i>	0.6
<i>alpha-3-mean</i>	0.69
<i>alpha-3-std</i>	0.63
<i>alpha-4-mean</i>	0.73
<i>alpha-4-std</i>	0.67
<i>alpha-5-mean</i>	0.73
<i>alpha-5-std</i>	0.66

<i>alpha-6-mean</i>	0.68
<i>alpha-6-std</i>	0.61
<i>alpha-7-mean</i>	0.79
<i>alpha-7-std</i>	0.77
<i>alpha-8-mean</i>	0.9
<i>alpha-8-std</i>	0.9
<i>sigma-0-mean</i>	0.62
<i>sigma-0-std</i>	0.62
<i>sigma-1-mean</i>	0.59
<i>sigma-1-std</i>	0.56
<i>sigma-2-mean</i>	0.63
<i>sigma-2-std</i>	0.58
<i>sigma-3-mean</i>	0.69
<i>sigma-3-std</i>	0.61
<i>sigma-4-mean</i>	0.73
<i>sigma-4-std</i>	0.64
<i>sigma-5-mean</i>	0.72
<i>sigma-5-std</i>	0.64
<i>sigma-6-mean</i>	0.68
<i>sigma-6-std</i>	0.61
<i>sigma-7-mean</i>	0.79
<i>sigma-7-std</i>	0.77
<i>sigma-8-mean</i>	0.89
<i>sigma-8-std</i>	0.9
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<i>beta-0-std</i>	0.61
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<i>beta-2-mean</i>	0.59
<i>beta-2-std</i>	0.59
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<i>beta-3-std</i>	0.6
<i>beta-4-mean</i>	0.68
<i>beta-4-std</i>	0.63
<i>beta-5-mean</i>	0.68

<i>beta-5-std</i>	0.63
<i>beta-6-mean</i>	0.65
<i>beta-6-std</i>	0.6
<i>beta-7-mean</i>	0.77
<i>beta-7-std</i>	0.75
<i>beta-8-mean</i>	0.89
<i>beta-8-std</i>	0.89
<i>gamma-0-mean</i>	0.54
<i>gamma-0-std</i>	0.59
<i>gamma-1-mean</i>	0.56
<i>gamma-1-std</i>	0.57
<i>gamma-2-mean</i>	0.64
<i>gamma-2-std</i>	0.6
<i>gamma-3-mean</i>	0.7
<i>gamma-3-std</i>	0.63
<i>gamma-4-mean</i>	0.74
<i>gamma-4-std</i>	0.67
<i>gamma-5-mean</i>	0.74
<i>gamma-5-std</i>	0.66
<i>gamma-6-mean</i>	0.71
<i>gamma-6-std</i>	0.64
<i>gamma-7-mean</i>	0.77
<i>gamma-7-std</i>	0.75
<i>gamma-8-mean</i>	0.89
<i>gamma-8-std</i>	0.89
<i>hypoxia_burden</i>	0.51
<i>t90burden</i>	0.38
<i>lmi</i>	0.53
<i>plmi</i>	0.51

Ablation Without Auxiliary Tasks

Methods: To assess the contribution of auxiliary physiological supervision, we trained an ablation model that removed all auxiliary heads (sleep staging and sleep-feature prediction) while keeping the encoder architecture, optimizer, and data splits identical. The resulting network had fewer parameters (158M vs. 199M) and received learning signal only from the high-level cognition and disease labels.

Results: Across all cohorts and outcomes, the no-auxiliary model showed minimal learning, with train, validation, and test performance remaining near chance for disease classification ($AUC \approx 0.50\text{--}0.55$) and near zero for cognition prediction ($r \approx 0.00\text{--}0.10$). In contrast, the full multi-task model exhibited consistent improvement across epochs and achieved substantially higher performance for both disease and cognition endpoints. The ablation thus demonstrated a categorical loss of predictive ability rather than a modest degradation.

Discussion: These findings indicate that, for the large multi-task architecture used here, high-level outcomes alone provide insufficient and sparse supervision to learn stable representations directly from raw sleep EEG. Because the model was originally designed with auxiliary physiological tasks in mind, removing these sources of dense, structured gradients leaves a 158M-parameter network effectively under-supervised, resulting in minimal learning. This does not preclude other, more compact or differently structured architectures from learning without auxiliary tasks, but it demonstrates that in our setting auxiliary supervision is essential for extracting meaningful physiological information from the signal.

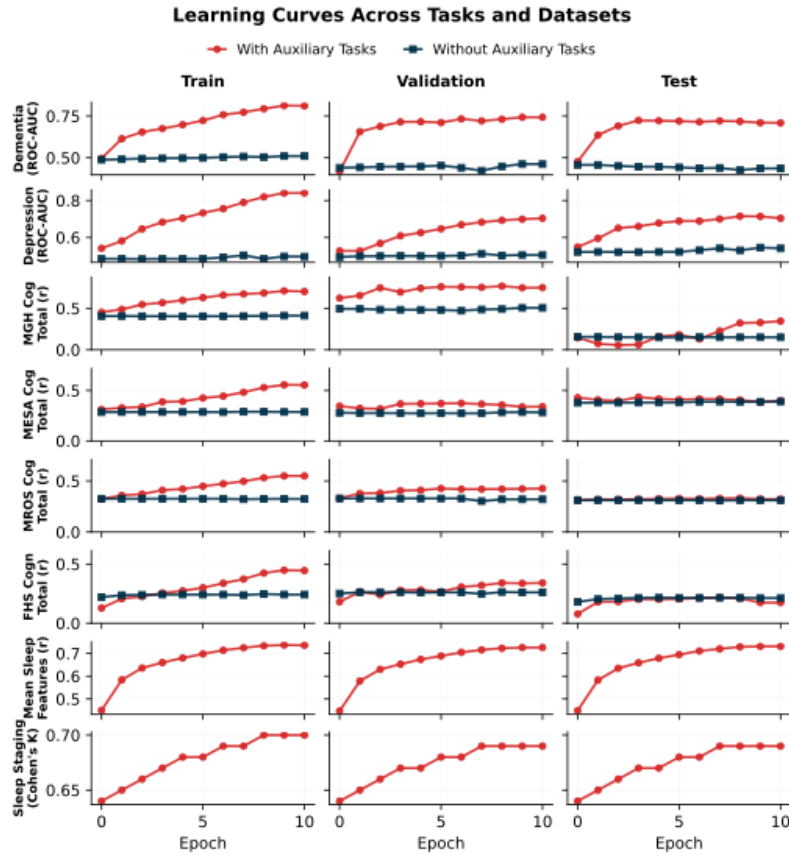


Figure S15. Learning curves for models trained with and without auxiliary tasks. Performance is shown across epochs for dementia, depression, and total cognition in six cohorts, separated by train, validation, and test sets. Models trained with auxiliary sleep-staging and sleep-feature prediction (red) show clear learning dynamics and strong generalization, whereas models without auxiliary tasks (black) remain flat and near chance across all outcomes. This illustrates that auxiliary physiological supervision is essential for learning meaningful representations from raw sleep EEG for our developed brain health model architecture.

Author Contributions:

WG served as the first author and led the project, overseeing data collection, software development, the design and training of deep learning models, and the execution of statistical analyses. Also prepared the initial full draft, generated the tables and figures, and coordinated the overall study workflow.

HS, AT, NT, PH, AG, KG, SN, YL, RT, and SK contributed to data acquisition, participated actively in group meetings, and provided critical feedback during the development process and manuscript revision.

SLH, KS, TH, SR, RA, and CS supported the data collection efforts and offered substantial input on refining the manuscript through rigorous review.

DM, HPL, RH, and EM offered mentorship, shaping the conceptual framework of the study and critically reviewing the manuscript.

SSC, RJT, and MBW, as co-senior authors, conceived the study, secured funding, and provided overarching leadership throughout the project. They oversaw all major aspects of data acquisition and analysis, offered strategic mentorship, and contributed in-depth critique during manuscript development.